Retail Analytics

Predicting Customer Purchase Behavior

MADE BY - ASHISH CHAMEL

COURSE - Capstone Project of Data Analytics and Generative AI

DATE OF SUBMISSON – 11-11-2025

PART 1 – EXCEL MODULE *(DATA CLEANING & PREPARATION)*

1.1 - Objective

The first step of the project was to understand and prepare the raw retail marketing dataset for further analysis.
Using **Microsoft Excel**, I focused on:

- Inspecting the dataset for missing or inconsistent data
- Cleaning and transforming fields for analysis
- Performing basic descriptive statistics
- Laying the foundation for segmentation and visualization in later stages

This phase ensured that all subsequent work (SQL, Python, Tableau) was based on clean, structured, and reliable data.

1.2 - Dataset Overview

The provided dataset contained detailed information about customer demographics, purchase behavior, and campaign response history.

Key columns included:

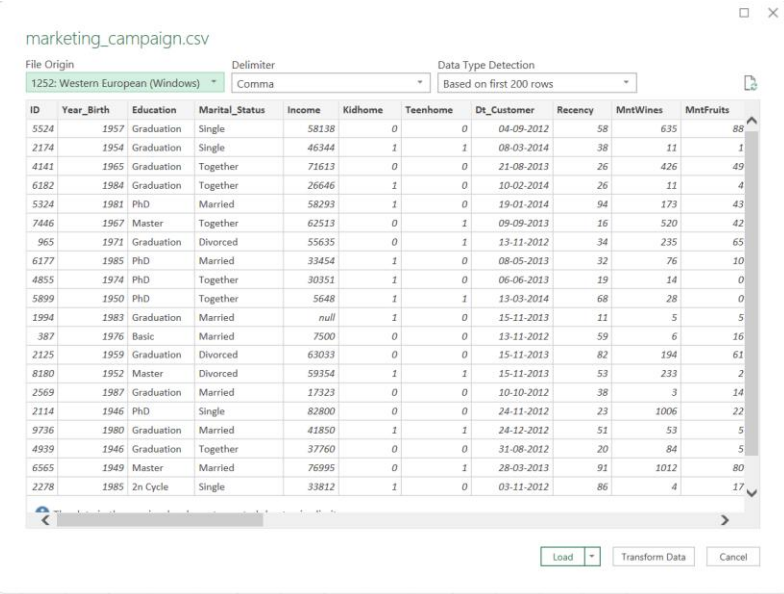
- ID – unique customer identifier
- Year_Birth – customer birth year
- Education, Marital_Status – demographic details
- Income – annual household income
- Kidhome, Teenhome – children information
- Recency – days since last purchase
- MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds – spending across product categories

- NumDealsPurchases, NumWebPurchases, NumStorePurchases – purchase channel activity
- Response – campaign response flag (target variable)

1.3 - Basic Cleaning

Step 1: Imported and Inspected the Dataset

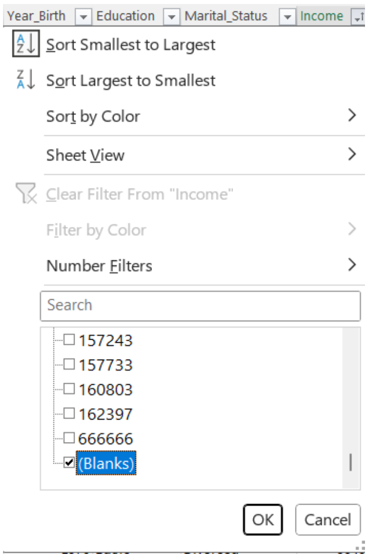
- I opened the provided .xlsx file in Excel and checked for data types, missing values, and irregularities.
- Verified there were no formatting breaks (e.g., blank columns, merged cells, trailing spaces).



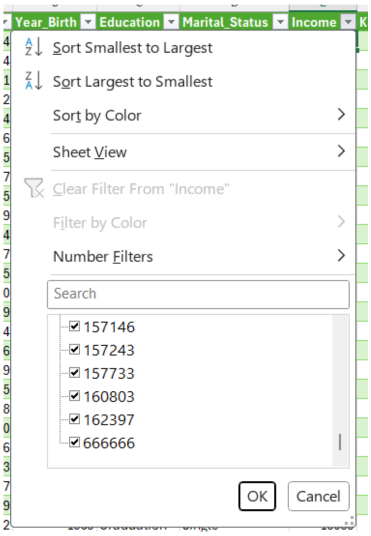
ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	MntFruits
5524	1957	Graduation	Single	58138	0	0	04-09-2012	58	635	88
2174	1954	Graduation	Single	46344	1	1	08-03-2014	38	11	1
4141	1965	Graduation	Together	71613	0	0	21-08-2013	26	426	49
6182	1984	Graduation	Together	26646	1	0	10-02-2014	26	11	4
5324	1981	PhD	Married	58293	1	0	19-01-2014	94	173	43
7446	1967	Master	Together	62513	0	1	09-09-2013	16	520	42
965	1971	Graduation	Divorced	55635	0	1	13-11-2012	34	235	65
6177	1985	PhD	Married	33454	1	0	08-05-2013	32	76	10
4855	1974	PhD	Together	30351	1	0	06-06-2013	19	14	0
5899	1950	PhD	Together	5648	1	1	13-03-2014	68	28	0
1994	1983	Graduation	Married	null	1	0	15-11-2013	11	5	5
387	1976	Basic	Married	7500	0	0	13-11-2012	59	6	16
2125	1959	Graduation	Divorced	63033	0	0	15-11-2013	82	194	61
8180	1952	Master	Divorced	59354	1	1	15-11-2013	53	233	2
2569	1987	Graduation	Married	17323	0	0	10-10-2012	38	3	14
2114	1946	PhD	Single	82800	0	0	24-11-2012	23	1006	22
9736	1980	Graduation	Married	41850	1	1	24-12-2012	51	53	5
4939	1946	Graduation	Together	37760	0	0	31-08-2012	20	84	5
6565	1949	Master	Married	76995	0	1	28-03-2013	91	1012	80
2278	1985	2n Cycle	Single	33812	1	0	03-11-2012	86	4	17

Step 2: Handled Missing and Null Values

- I identified missing or null entries in the **Income** column and **Education** column.



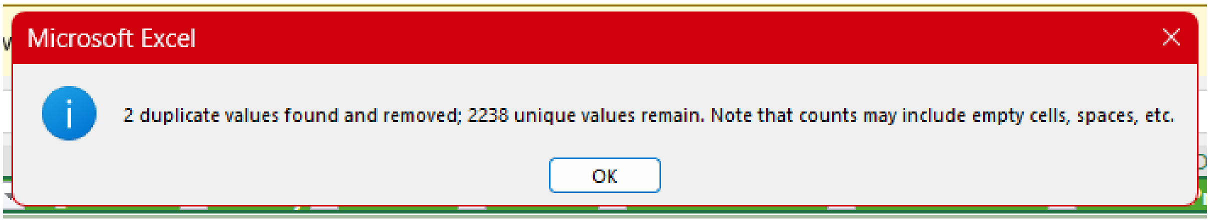
- Replaced missing numeric values with the **median** (since it's less affected by outliers).



- Same treatment for categorical nulls, filled with **"Unknown"**.

Step 3: Removed Duplicates and Invalid Rows

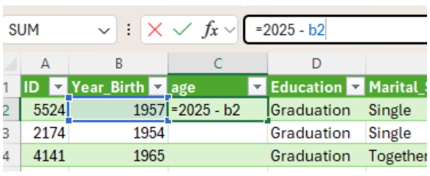
- Used *Data → Remove Duplicates* on the ID column to eliminate duplicate customer entries.

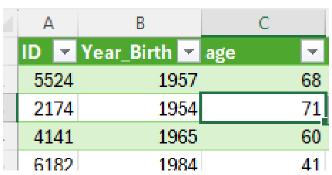


- Filtered out invalid records where:
  - Income ≤ 0
  - Year_Birth outside realistic range (1900–2000)
- This ensured a realistic and consistent dataset.

Step 4: Created New Calculated Columns

- Added a derived **Age** column:
- Age = 2024 - Year_Birth





- Created a **Total Spending** column:
- Total_Spend = MntWines + MntFruits + MntMeatProducts + MntFishProducts + MntSweetProducts + MntGoldProds
- This made customer segmentation and RFM scoring much easier later.

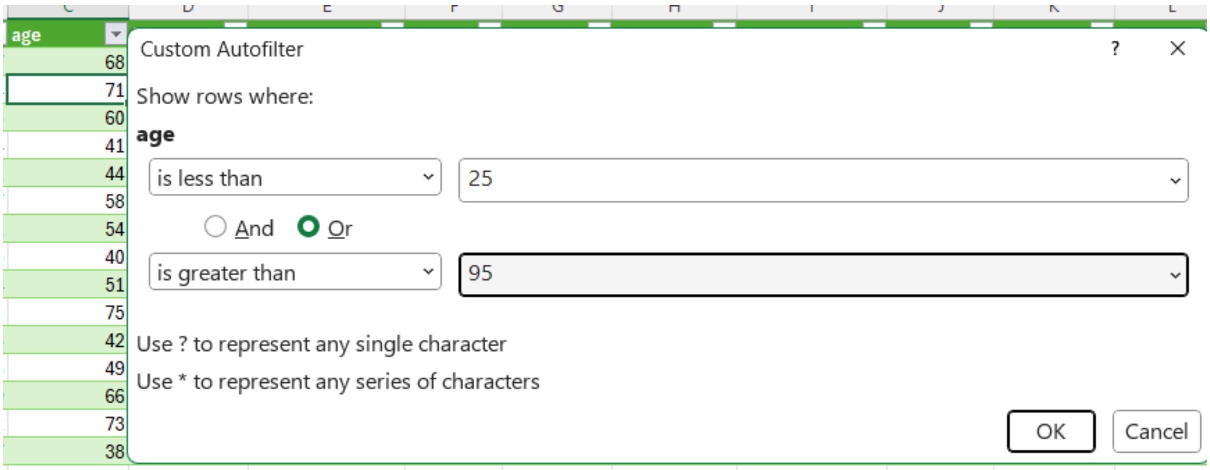
Step 5: Filtering Invalid Entries

While scanning the **Income** column, I discovered that a few records contained either missing, negative, or zero values, which were logically invalid.
In addition, some outliers had extremely high incomes that didn't match typical customer ranges.

To correct this:

- I applied filters and manually reviewed extreme income values.
- Removed rows with zero or unrealistic income figures.
- Retained high-income entries only when supported by consistent spending patterns.

This manual validation ensured that the cleaned dataset remained realistic and aligned with expected customer demographics.



1.4 - Task 1 - Statistical Summary and Initial Data Insights

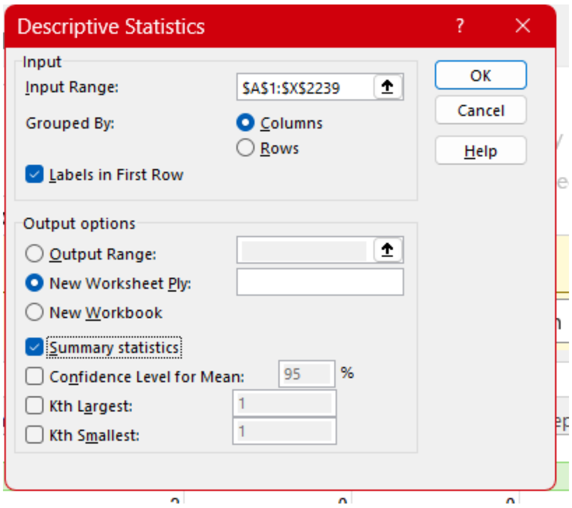
To compute descriptive statistics for all numerical variables in the dataset (e.g., Income, Recency, Frequency, Monetary, Mnt* columns) and derive preliminary insights into customer demographics, spending patterns, and campaign responses.

Step 1 Select the Numeric Range
Opened the raw dataset and selected all numeric columns from Year_Birth to Response using the range B1:AB2240.
Ensured the first row contained headers.

Step 2 Run the Descriptive Statistics Tool

- Go to Data > Data Analysis > Descriptive Statistics

- Input Range: raw!B1:AB2240
- Grouped By: Columns
- Labels in First Row:
- Output Options: New Worksheet Ply
- Summary Statistics:
  Clicked OK to generate results.



Step 3 Create a Cleaned Summary Table

- Copied results for Year_Birth → Response
- Pasted into a new sheet named Cleaned_Summary
- Retained only key metrics: Mean, Standard Deviation, Min, Median, Max, Count
- Deleted less-useful rows (Skewness, Kurtosis, Range, etc.)

	A	B	C	D	E	F	G	H	I	
1	*Year_Birth*		*age*		*Income*		*Kidhome*		*Teenhome*	
2										
3	Mean	1968.902	Mean	56.09835	Mean	52227.41	Mean	0.444345	Mean	0.50
4	Standard E	0.247414	Standard E	0.247414	Standard E	529.4903	Standard E	0.011385	Standard E	0.01
5	Median	1970	Median	55	Median	51382	Median	0	Median	
6	Mode	1976	Mode	49	Mode	51382	Mode	0	Mode	
7	Standard D	11.70192	Standard D	11.70192	Standard D	25043.27	Standard D	0.538467	Standard D	0.54
8	Sample Va	136.9349	Sample Va	136.9349	Sample Va	6.27E+08	Sample Va	0.289947	Sample Va	0.29
9	Kurtosis	-0.79583	Kurtosis	-0.79583	Kurtosis	161.492	Kurtosis	-0.77944	Kurtosis	-(
10	Skewness	-0.09327	Skewness	0.093266	Skewness	6.806236	Skewness	0.635078	Skewness	0.40
11	Range	56	Range	56	Range	664936	Range	2	Range	
12	Minimum	1940	Minimum	29	Minimum	1730	Minimum	0	Minimum	
13	Maximum	1996	Maximum	85	Maximum	666666	Maximum	2	Maximum	
14	Sum	4404433	Sum	125492	Sum	1.17E+08	Sum	994	Sum	
15	Count	2237	Count	2237	Count	2237	Count	2237	Count	

Step 4 - Transpose for Readability

Copied selected rows (Mean → Count), then:

- Right-click > Paste Special > Transpose

Step 5 Formatting

- Bold + center-align headers
- Format numeric columns (Comma Style with 2 decimals)
- Add borders and adjust column widths

Result:
A compact, easy-to-read statistical summary sheet showing descriptive metrics for all numeric features.

Feature	Year_Birth	age	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts
Mean	1968.901654	56.098346	52227.41305	0.444345105	0.506481895	49.10460438	303.9955297	26.2704515	166.9168529	37.5230219	27.0688422
Median	1970	55	51382	0	0	49	174	8	67	12	8
Mode	1976	49	51382	0	0	56	2	0	7	0	0
Standard Deviation	11.70191726	11.70191726	25043.26665	0.538467051	0.544592876	28.9560731	336.5743823	39.71597157	225.6611579	54.63990947	41.29394931
Range	56	56	664936	2	2	99	1493	199	1725	259	263
Minimum	1940	29	1730	0	0	0	0	0	0	0	0
Maximum	1996	85	666666	2	2	99	1493	199	1725	259	263
Sum	4404433	125492	116832723	994	1133	109847	680038	58767	373393	83939	60553
Count	2237	2237	2237	2237	2237	2237	2237	2237	2237	2237	2237

1.4.1 - Key Observations and Insights

1. Customer Demographics

- Average Age: ≈ 56 years (29–85 range) → mostly middle-aged to older adults → stable, loyal base.
- Year of Birth: centered around 1969 → mature audience with purchasing power.

2. Income Distribution

- Mean Income: ≈ ₹ 52 K; Std Dev ≈ ₹ 25 K; Range = ₹ 1.7 K – ₹ 666 K.
  → Large disparity → a few high-income outliers (skew to right).
  → Median ≈ ₹ 51 K → majority middle-income customers.

3. Household Composition

- Kidhome (0.44) and Teenhome (0.50) → most households have 0–1 children → small-family or empty-nest segments.

4. Recency (Engagement Metric)

- Mean Recency = 49 days (range 0–99) → average purchase ~1.5 months ago.
  → Moderate engagement; some inactive customers present → re-activation opportunity.

5. Spending Behavior (Monetary Variables)

- Wine = 304, Meat = 167, Gold = 44 (Mean Values)
  → Premium preferences — affluent lifestyle segment.
  → Fruits, Fish, Sweet < 50 → secondary categories.
  → High std devs in Wines/Meat/Gold → wide spending diversity.

6. Purchase Channels

- NumDealsPurchases = 2.3, NumStorePurchases = 5.8 → in-store dominates.
- NumWebPurchases = 4.1, NumCatalogPurchases = 2.6 → online growing but secondary.
- Web Visits ≈ 5 / month → strong digital potential with low conversion gap.

7. Campaign Responses

- AcceptedCmp1–5 ≈ 7 %, Response ≈ 15 % → recent campaigns performing better → positive trend in customer engagement.

8. Complaints

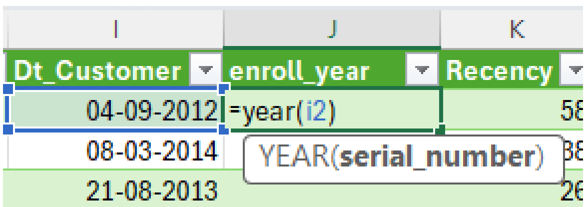
- Complain ≈ 1 % → very low dissatisfaction → solid service quality.

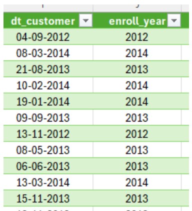
1.5 -Task 2 - Line Chart: Customer Enrolments by Year

To visualize customer acquisition trends over time and identify any significant peaks or declines in new enrolments. This helps the RetailVC team understand how marketing campaigns, promotions, and external factors have influenced customer growth year by year.

Step 1 - Create the "Enroll_Year" column
In the raw dataset, a column named Dt_Customer captured the date each customer joined. To extract the joining year:

1. Insert a new column beside Dt_Customer and name it Enroll_Year.
2. In the first data row, type the formula:
3. =YEAR(I2)





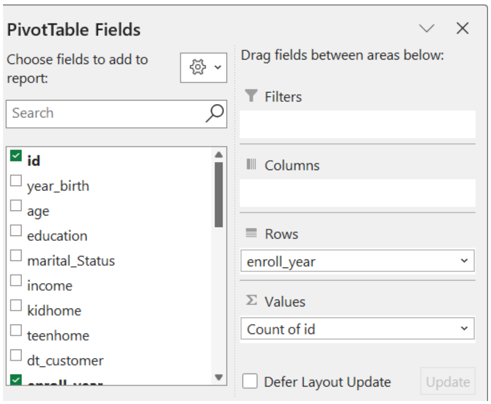
dt_customer	enroll_year
04-09-2012	2012
08-03-2014	2014
21-08-2013	2013
10-02-2014	2014
19-01-2014	2014
09-09-2013	2013
13-11-2012	2012
08-05-2013	2013
06-06-2013	2013
13-03-2014	2014
15-11-2013	2013

4. Press Enter, then drag the fill handle down to copy the formula for all rows.
   This converts each full registration date into its corresponding year (e.g., 2012, 2013, 2014).

Step 2 -Build the Pivot Table

1. Select the full dataset (including your new Enroll_Year column).

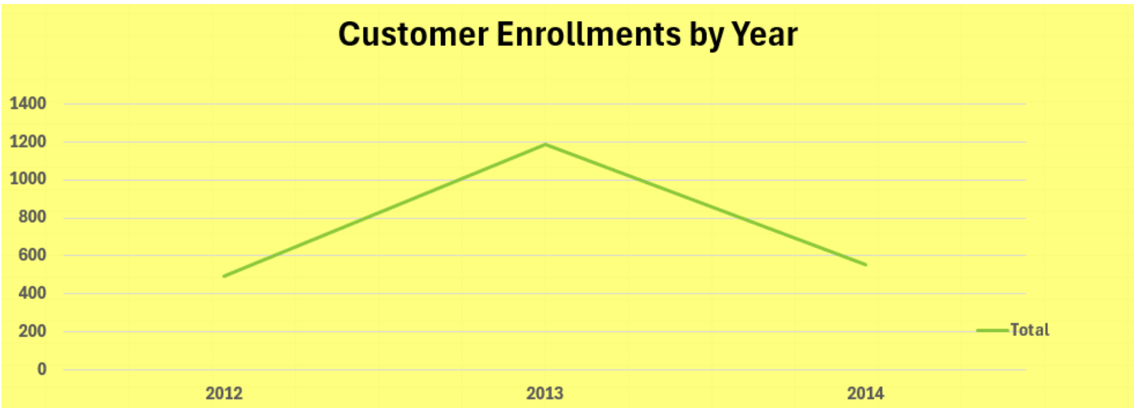
2. Go to Insert → PivotTable → From Table/Range.
3. In the PivotTable Field List:
    - Drag Enroll_Year → Rows
    - Drag ID (Customer ID) → Values
    - Right-click on the value field → Value Field Settings → Count (so it counts customers instead of summing).



4. Rename the value field as Number of Customers for clarity.

Step 3 Insert the Line Chart

1. Click anywhere inside the pivot.
2. Go to Insert → Line → Line with Markers.
3. Add chart elements:
    - Chart Title : "Customer Enrolments by Year"
    - Horizontal Axis Title : "Year of Enrolment"
    - Vertical Axis Title : "Number of Customers"
4. Format the line for readability
5. Ensure axis years are in ascending order



1.5.1 - Interpretation:

- Enrollments increased sharply in 2013, reaching 1,187 customers, compared to 494 in 2012.
- A decline occurred in 2014 (556 customers), following the strong growth of 2013.
- The peak in 2013 indicates a period of high marketing effectiveness or successful promotional campaigns.
- The drop in 2014 could suggest reduced campaign intensity, market saturation, or customer fatigue.
- RetailVC should analyze the drivers behind the 2013 success and replicate or adapt those strategies for future acquisition efforts.
- Overall, the trend reflects rapid growth followed by stabilization, signaling a need for renewed engagement to sustain momentum.

1.6 - Task 3 : Cross-Tabulated Analysis: Marketing Response vs Education Level

To analyze how customers with different education levels responded to marketing campaigns. This cross-tab helps identify whether education is a key factor influencing engagement and response likelihood.

Step 1 Create the Pivot Table

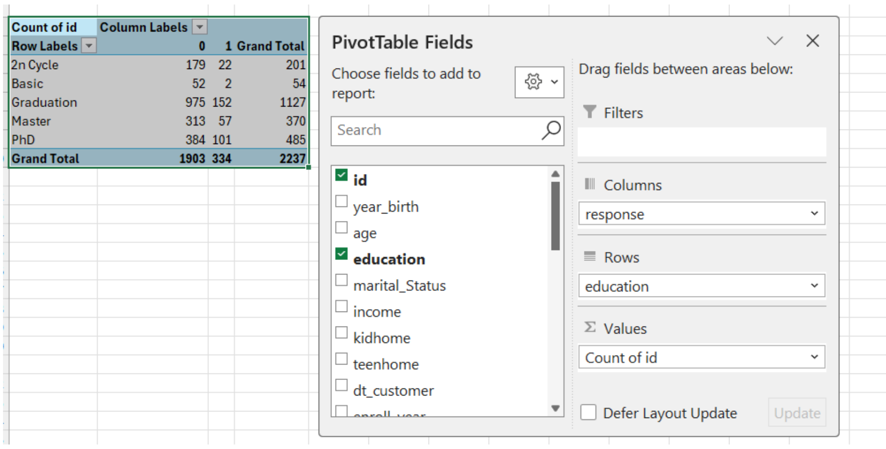
1. Open the cleaned dataset (with Response, Education, and ID columns).
2. Select the entire range (Ctrl + A).
3. Go to Insert → PivotTable → From Table/Range.
4. Choose *New Worksheet* and click OK.

Step 2 Configure Pivot Fields

In the PivotTable Fields pane:

- Drag Education → *Rows*
- Drag Response → *Columns*
- Drag ID → *Values*

Count of id	Column Labels		
Row Labels	**0**	**1**	**Grand Total**
2n Cycle	179	22	201
Basic	52	2	54
Graduation	975	152	1127
Master	313	57	370
PhD	384	101	485
Grand Total	**1903**	**334**	**2237**

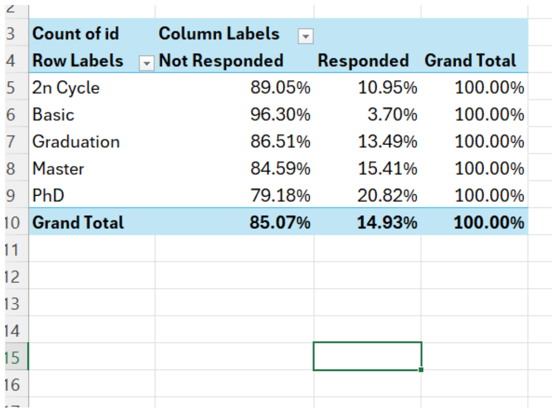


Count of id	Column Labels		
Row Labels	0	1	Grand Total
2n Cycle	179	22	201
Basic	52	2	54
Graduation	975	152	1127
Master	313	57	370
PhD	384	101	485
Grand Total	1903	334	2237

It should automatically show Count of ID — representing the number of customers per education level who did or didn't respond.
If not, right-click → Value Field Settings → Count.

Step 3 Convert Counts to % of Row Total
This allows you to compare within each education group:

1. Right-click any number inside the PivotTable → Show Values As → % of Row Total.
2. Format the percentages: right-click → Number Format → Percentage (1 decimal place).
   Now, for each education level, you'll see what proportion responded "1" (Yes) versus "0" (No).



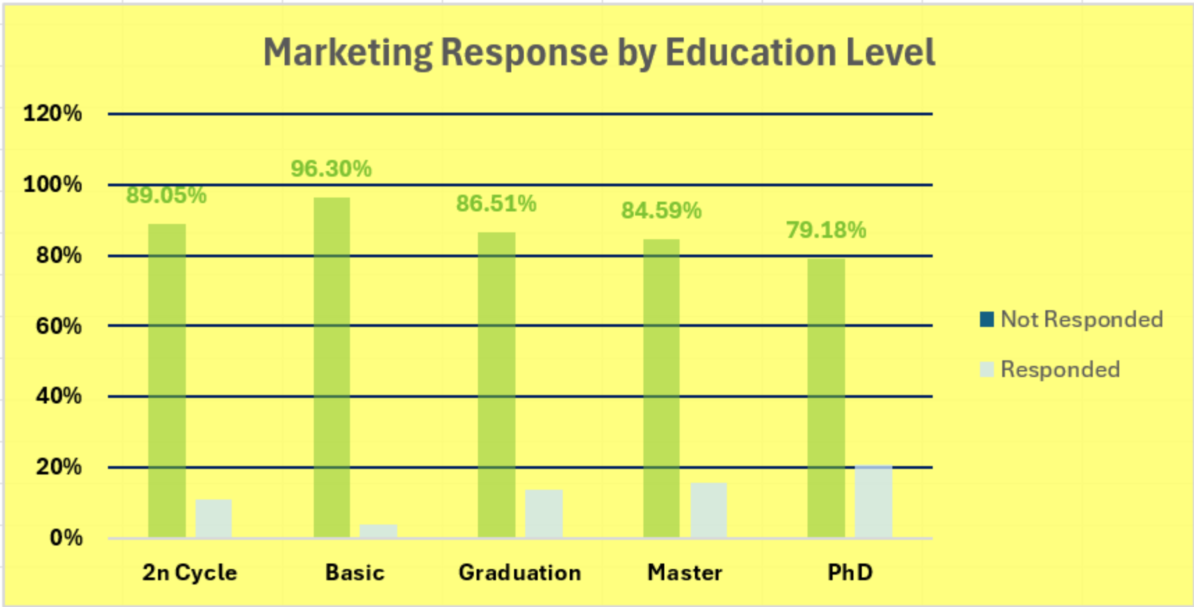
Count of id	Column Labels		
Row Labels	Not Responded	Responded	Grand Total
2n Cycle	89.05%	10.95%	100.00%
Basic	96.30%	3.70%	100.00%
Graduation	86.51%	13.49%	100.00%
Master	84.59%	15.41%	100.00%
PhD	79.18%	20.82%	100.00%
Grand Total	85.07%	14.93%	100.00%

Step 4 Insert the Visualization

1. Click anywhere on the PivotTable.
2. Go to Insert → Charts → Clustered Column.
3. Add chart elements:
   - Chart Title: *Marketing Response by Education Level*
   - X-Axis: Education Level

- Y-Axis: % of Respondents

4. Format for clarity:



1.6.1 - Interpretation:

Customers with PhD (20.8%) and Master (15.4%) education levels show the highest marketing response rates.

- Graduates (13.5%) also respond moderately well, indicating a consistent pattern among higher education groups.
- Those with Basic (3.7%) or 2n Cycle (10.9%) education show low responsiveness, suggesting limited campaign effectiveness in these segments.
- This pattern implies that education level is positively associated with marketing engagement.
- RetailVC can prioritize Graduate–PhD customers for targeted campaigns and refine messaging for lower-education groups using simpler communication or offline outreach.

1.7 - Task 4 : Boxplot Analysis of Customer Income Distribution

To analyze the spread and variability of customer income, detect outliers, and understand how income diversity might influence purchasing behavior.
This helps RetailVC identify distinct customer segments from low-income to high-spending premium customers.

Step 1 Select the Income Data

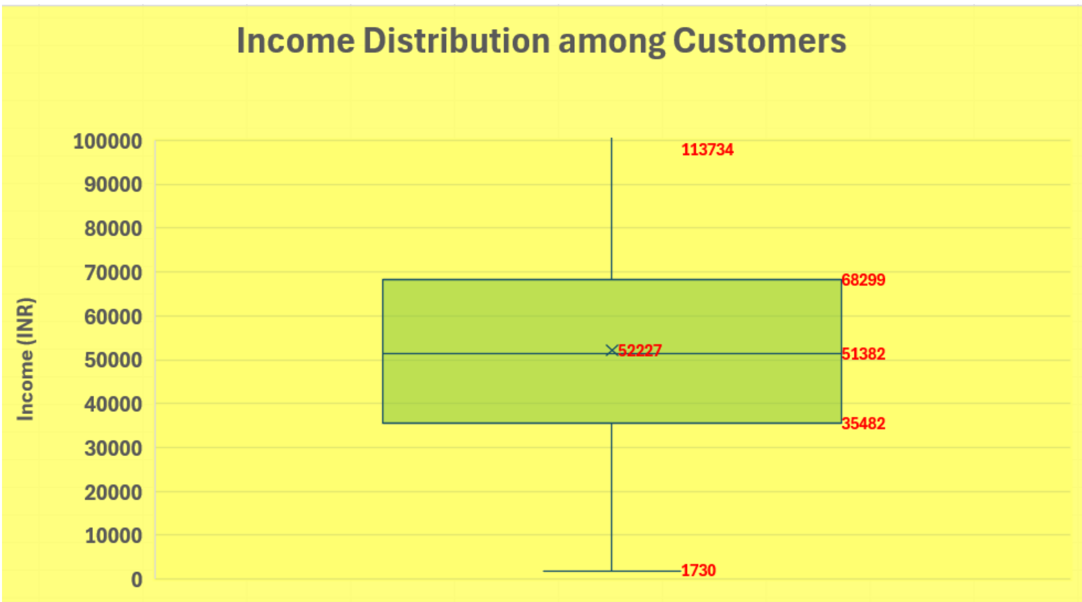
1. Open your raw or cleaned dataset.
2. Click on the Income column header (numeric values only).
3. Ensure no empty rows or non-numeric entries are selected.
   This isolates the target variable for visual distribution analysis.

Step 2 — Insert a Box & Whisker Chart From the top ribbon, go to:
Insert → Statistical Chart → Box & Whisker
(If your Excel version doesn't display this directly, go to:
Insert → All Charts → Box & Whisker *).*

1. Excel automatically generates a boxplot representing the Income column.

Step 3 — Format the Chart for Readability

1. Click the chart once to activate it.
2. Open Chart Elements (+ icon) → enable:
    - Axis Titles
    - Chart Title
3. Rename the title as:
   "Income Distribution among Customers"
4. Add a Y-axis title:
   "Income (₹)"
5. Format the design:
    - Choose a soft fill color for the box
    - Set axis limits to focus on the realistic income range:
        - Minimum: 0
        - Maximum: 100000 (or slightly above your 75th percentile).



1.7.1 - Key Observations and Insights

- The median customer income is approximately ₹51,382 — indicating that the dataset's income center lies in the mid-tier range.
- The interquartile range (IQR) — ₹35,482 to ₹68,299 — shows a moderately wide spread, reflecting income diversity among customers.
- The mean (₹52,227) is close to the median, suggesting a nearly symmetric distribution with mild right skew due to a few high-income outliers.
- Minimum income (~₹1,730) and maximum (~₹113,734) confirm that the dataset includes both lower-income and high-income individuals.
- The presence of upper-end outliers (dots above the whiskers) points to a small segment of premium customers, crucial for high-value targeting.

1.8 - Task 5 Customer Age Calculation and Distribution (Histogram Analysis)

To calculate the age of each customer using the Year of Birth field and visualize the age distribution using a histogram.
This analysis helps identify dominant customer age segments and guides targeted marketing strategies for RetailVC.

Step-by-Step Process

Step 1 Calculate the Age of Customers

1. In your cleaned dataset, ensure there is a column named Year_Birth.
2. Insert a new column beside it and label it Age.
3. In the first row of this new column (e.g., C2), type the formula:
4. =YEAR(TODAY()) - B2

(Assuming Year_Birth is in column B.)

5. Press Enter and drag down to fill the formula for all customers.
 This will calculate each customer's age as of the current year.

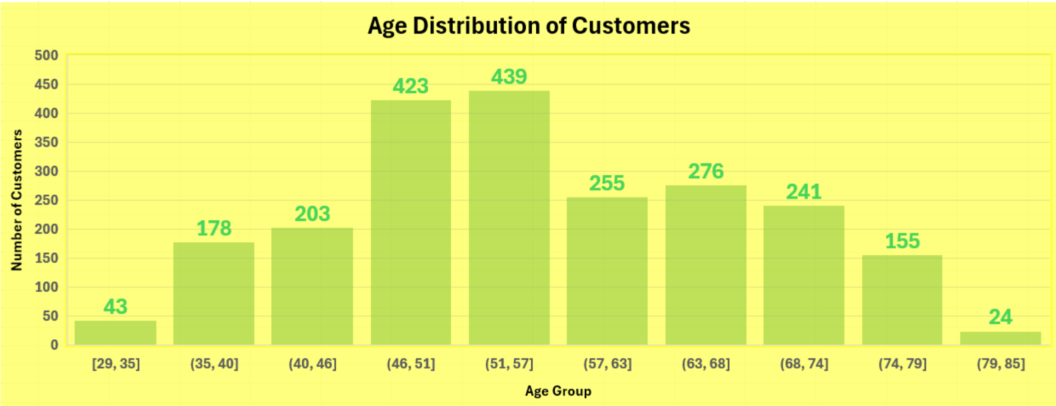
year_birth	age
1957	=2025 - B2
1954	=2025 - B3
1965	=2025 - B4
1984	=2025 - B5
1981	=2025 - B6
1967	=2025 - B7
1971	=2025 - B8
1985	=2025 - B9
1974	=2025 - B10
1950	=2025 - B11
1983	=2025 - B12
1976	=2025 - B13
1959	=2025 - B14

Step 2 Create the Histogram Visualization

1. Select the Age column (numeric values only, no header).
2. Go to Insert → Statistical Chart → Histogram.
3. Excel will automatically group the ages into bins and display a frequency distribution chart showing how many customers fall into each age range.

Step 3 Refine the Chart for Presentation

1. Click on the chart → enable Chart Elements (+ icon) → tick:
   - Axis Titles
   - Chart Title
2. Customize titles:
   - Chart Title: "Age Distribution of Customers"
   - X-Axis Title: "Age Group"
   - Y-Axis Title: "Number of Customers"
3. Adjust the bin width for readability:
   - Right-click on the X-axis → Format Axis → Bin Width = 5 (or 10, depending on spread).
     → 5-year bins usually give a clear, balanced view (e.g., 25–29, 30–34, 35–39, etc.).
4. Format styling for professionalism:



<u>1.8.1 - Results and Interpretation</u>

- The majority of RetailVC customers fall between 46–57 years, representing the brand's primary target demographic middle-aged adults with stable income and purchasing power.
- There's a clear decline below age 35 and above 70, highlighting limited engagement among younger and older segments.

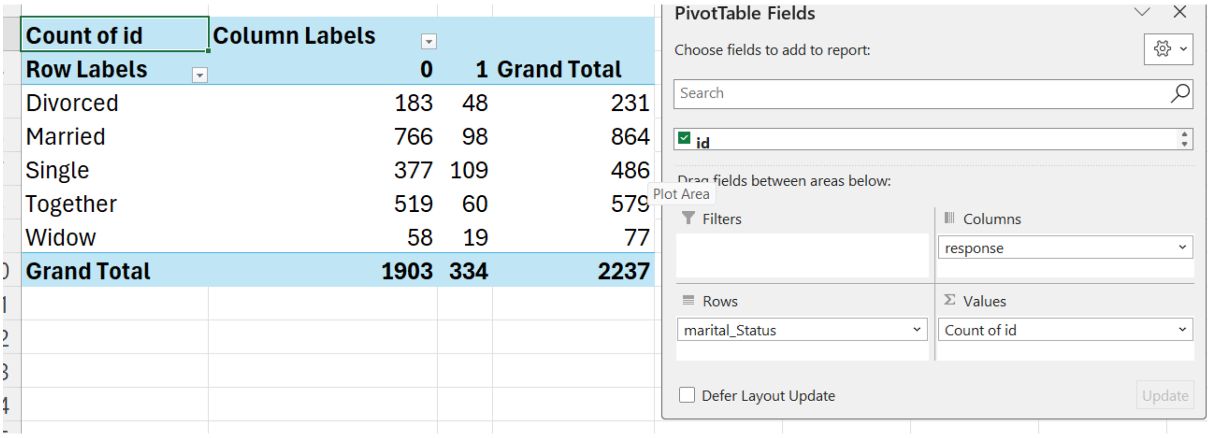
- The distribution is slightly left-skewed, indicating that most customers are concentrated in older age brackets.
- Combined, the 46–63 age range accounts for over 50% of all customers a strong mid-age dominance.

1.9 - Task 6 Marketing Response by Marital Status

To evaluate how customers' marital status influences their likelihood of responding to marketing campaigns.
By examining response patterns across different relationship categories, RetailVC can better tailor communication and promotional offers to each demographic.

Step 1 Create the Pivot Table

1. Select the entire dataset (ensure it includes Marital_Status, Response, and ID or Customer columns).
2. Go to Insert → PivotTable → From Table/Range.
3. Choose *New Worksheet* and click OK.
4. Configure fields in the PivotTable pane:
    - Rows → Marital_Status
    - Columns → Response
    - Values → Count of ID (or Count of Customer)



Count of id	Column Labels		
Row Labels	0	1	Grand Total
Divorced	183	48	231
Married	766	98	864
Single	377	109	486
Together	519	60	579
Widow	58	19	77
Grand Total	1903	334	2237

Step 2 Convert Counts to Percentages

1. Right-click any number in the PivotTable → Show Values As → % of Row Total.
2. Format as Percentage (1 decimal place).

Step 3 Insert a 100% Stacked Bar Chart

1. Highlight the entire PivotTable (excluding the Grand Total row).
2. Go to Insert → Bar/Column Chart → 100% Stacked Bar.

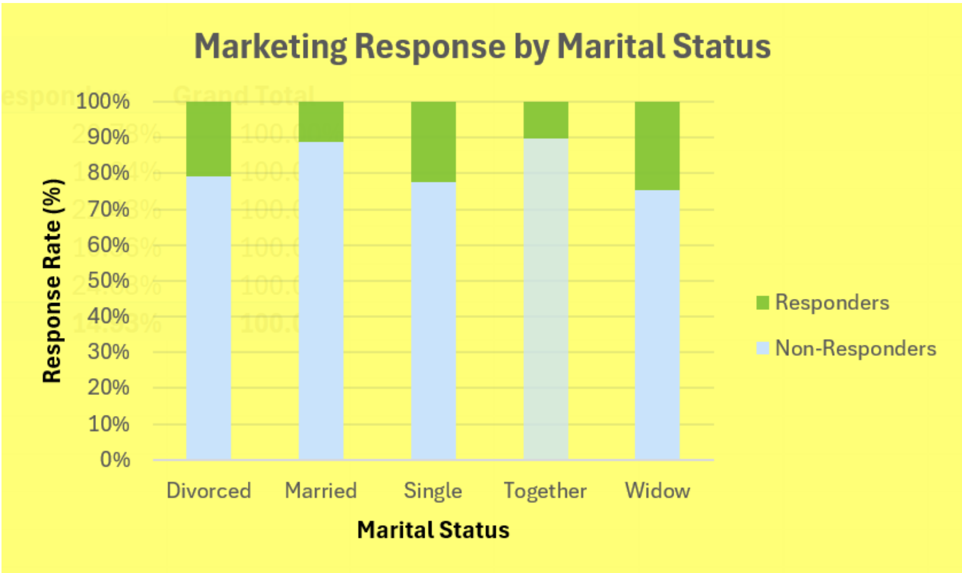
3. Rename the chart title to:
   "Marketing Response by Marital Status".

4. Adjust axes:

   - X-Axis Title: "Response Rate (%)"
   - Y-Axis Title: "Marital Status"

Step 4 Format the Chart for a Professional Look

- Response = 1: Fill bar color = Green (indicates successful response).
- Response = 0: Fill bar color = Gray or light red (non-response).
- Add data labels → percentages displayed directly on bars.
- Keep the background clean — white or light beige for presentation.
- Optional: Reorder the marital categories (e.g., Single → Together → Married → Divorced → Widow) for logical storytelling.



1.9.1 - Key Findings and Interpretation

- Single customers show the highest response rate (≈18%), indicating they are the most receptive to promotional campaigns.
- Customers living together ("Together") also engage well, slightly above the divorced and widowed groups.
- Married customers, though forming the largest overall segment, show the lowest conversion rate (~10%), suggesting either more cautious spending or reduced interest in promotional campaigns.

- The pattern reveals that relationship status significantly affects campaign performance, aligning with broader marketing trends — younger, single, or cohabiting individuals are more open to offers than stable, married households.

(A) - Key Insights from the Excel Module

1. Customer Demographics

- The average customer age is ~56 years, with most falling between 46–57 years — a mature, stable, and financially capable segment.
- Year of Birth (1969–1970) aligns with mid-aged professionals who value reliability and brand reputation.
- Household Composition: Majority have 0–1 children, confirming that RetailVC's base consists primarily of small-family or empty-nest households.

Implication: Marketing messages should emphasize quality, lifestyle enhancement, and convenience over price competition.

2. Income Distribution and Customer Segments

- Mean Income: ₹52,227 | IQR: ₹35k–₹68k
- Income distribution is moderately right-skewed, with a few high-income outliers driving the mean upward.
- The dataset spans low to high earners, signaling a diverse customer base across multiple economic segments.

Implication:

- Mid-income group → core market (discount-driven).
- High-income outliers → premium product opportunities (wine, gold, meat).

3. Enrolment Trend Analysis

- Customer enrolments spiked sharply in 2013 (1,187) - double 2012 - then fell to 556 in 2014.
- This indicates that 2013 campaigns were exceptionally effective, but momentum wasn't sustained.

Implication:

- Review and replicate 2013 campaign tactics (timing, channel mix, offers).
- Analyze 2014 drop for causes (budget cuts, fatigue, competition).

4. Education and Response Behavior

- PhD (20.8%) and Master's (15.4%) customers had the highest campaign acceptance rates.

- Engagement drops progressively with lower education levels (Basic = 3.7%).
- Indicates a positive correlation between education level and responsiveness.

Implication:

- Use data-driven and value-centric messaging for educated customers.
- Simplify campaign visuals and language for less-educated segments to increase reach.

5. Age-Based Purchase Behavior

- Age distribution forms a bell-like shape, peaking in 46–57 years range.
- Younger (<35) and older (>70) segments are smaller and less engaged.
- Indicates that RetailVC appeals primarily to mid-life, career-stable consumers.

Implication:

- Focus campaigns on 45–60 age group (core market).
- Develop dedicated strategies for youth onboarding (digital-first, gamified offers).
- Explore senior loyalty programs for retention.

6. Marital Status and Response Patterns

- Single (18%) and Together (16%) customers responded most positively.
- Married customers, though forming the largest base, showed lowest conversion (~10%).
- Relationship status clearly influences responsiveness — singles/cohabiting = opportunity segment.

Implication:

- Singles: use lifestyle-based offers (flexible pricing, adventure/experience themes).
- Married: focus on value-for-money bundles and family deals.
- Divorced/Widowed: personalized engagement or community-driven messaging.

---

7. Channel and Purchase Behavior

- In-store purchases (5.8 avg) still dominate, but web purchases (4.1) are rising.
- Web visits (≈5/month) show digital curiosity — but conversion lag remains.

Implication:

- Strengthen online conversion funnels (retargeting, exclusive web discounts).

- Integrate omnichannel offers to bridge store and digital behavior.

8. Campaign Effectiveness

- Average Response Rate: 15% (up from ~7% in earlier campaigns).
- Indicates improvement in campaign quality and targeting precision.
- Yet overall acceptance remains modest — implying room for optimization.

Implication:

- Use behavioral segmentation (based on education, income, and marital status) for next campaigns.
- Focus on message timing and channel personalization (SMS vs email vs in-store).

2 - SQL Module

The SQL module focused on structuring, cleaning, and exploring the RetailVC marketing dataset using MySQL.
It involved creating the database schema, importing data through the wizard, and performing key analytical queries.
This stage ensured data integrity and built a reliable foundation for deeper analysis in Python and Tableau.

2.1 - Task 1 - Create Database and Schema

Set up the SQL environment for the *RetailVC* project by creating a new database, defining its schema, and preparing it for data import.

1. Open MySQL Workbench and connect to the local server.
2. Enable local_infile to allow data imports:
→ Go to *Edit* → *Preferences* → *SQL Editor* → check "Allow LOAD DATA LOCAL INFILE".
→ Restart the connection.
3. Create a new database named retailvc and set it as your working schema:

    CREATE DATABASE IF NOT EXISTS retailvc

    CHARACTER SET utf8mb4

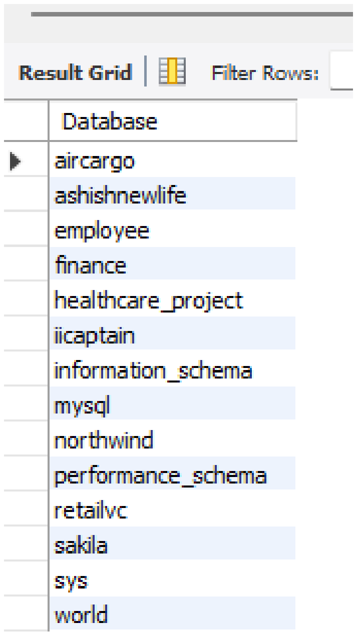
    COLLATE utf8mb4_general_ci;

    USE retailvc;

```
-- Task 1: Create Database and Schema for RetailVC

-- 1️⃣Create the main database
CREATE DATABASE IF NOT EXISTS retailvc
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_general_ci;

-- 2️⃣Use the database
USE retailvc;

-- 3️⃣Confirm the active schema
SELECT DATABASE() AS current_schema;

-- 4️⃣Optional: view all available databases
SHOW DATABASES;

```

4. Confirm the active schema:

    SHOW DATABASES;

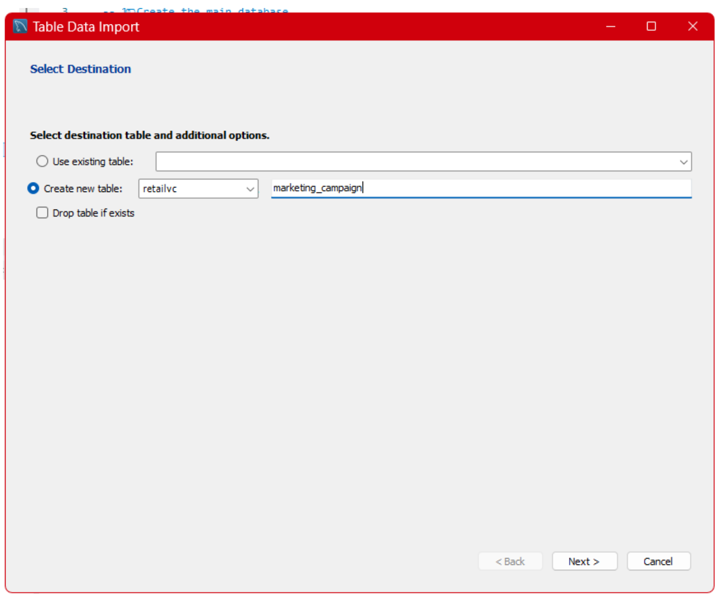


5. Refresh the *Schemas* panel to verify that retailvc is visible.

2.2 - Task 2 – Data Import (MySQL Workbench Wizard)

Import the retail marketing dataset into MySQL efficiently using the *Table Data Import Wizard*.

1. In *Schemas* panel → right-click retailvc → choose Table Data Import Wizard.
2. Browse for the dataset:
marketing_campaign_clean.csv
3. Choose Import to New Table, name it marketing_campaign.

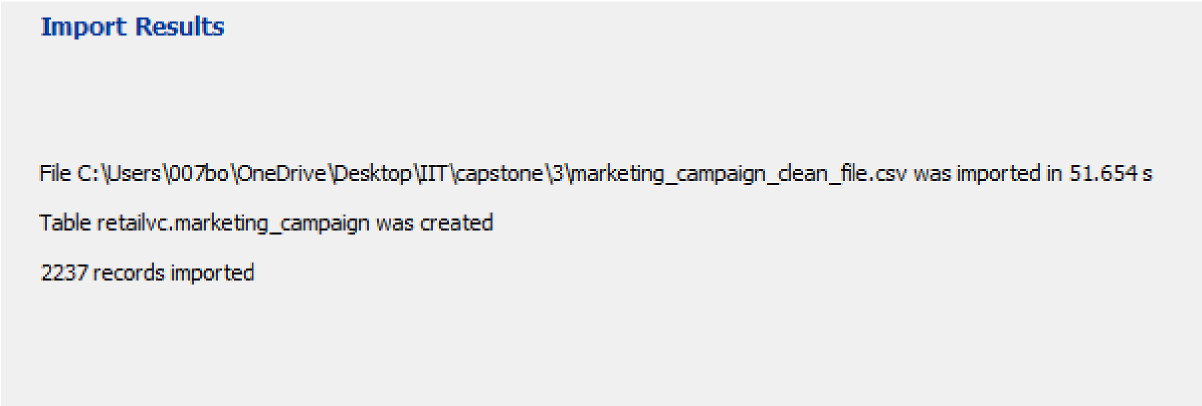


4. Let the wizard auto-map columns (text → VARCHAR, numeric → DOUBLE).

5. Click Next → Finish, then refresh schema to confirm import.

Result

Table marketing_campaign created successfully with all columns properly recognized.
The imported dataset contains 2,240 rows and 22 columns, identical to the raw data file.

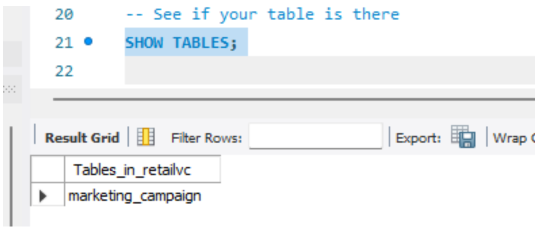


2.3 - Task 3 - Verify and Clean Imported Data

Ensure the imported data is valid, consistent, and free from duplicates or missing values.

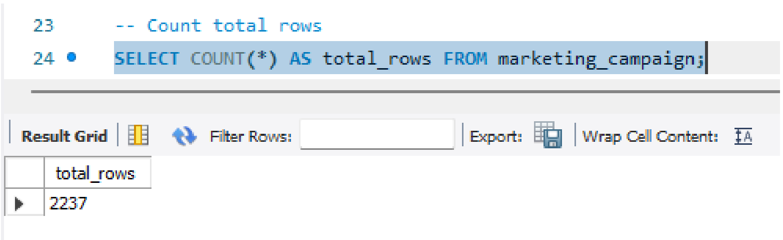
Step 1 – Confirm Table and Row Count

SHOW TABLES;



SELECT COUNT(*) AS total_rows FROM marketing_campaign;

→ Confirms all 2,237 rows successfully imported.



Step 2 – Data Preview

SELECT * FROM marketing_campaign LIMIT 10;

→ Ensures column alignment and proper headers.

```sql
-- Step 2. Quick peek at your data
SELECT * FROM marketing_campaign LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

ï»¿id	year_birth	age	education	marital_Status	income	kidhome	teenhome	dt_customer	enroll_year	recency	mnt_wines	mnt_fruits	mnt__eat_products	mnt_fish_products	mnt_sweet_
5524	1957	68	Graduation	Single	58138	0	0	04-09-2012	2012	58	635	88	546	172	88
2174	1954	71	Graduation	Single	46344	1	1	08-03-2014	2014	38	11	1	6	2	1
4141	1965	60	Graduation	Together	71613	0	0	21-08-2013	2013	26	426	49	127	111	21
6182	1984	41	Graduation	Together	26646	1	0	10-02-2014	2014	26	11	4	20	10	3
5324	1981	44	PhD	Married	58293	1	0	19-01-2014	2014	94	173	43	118	46	27
7446	1967	58	Master	Together	62513	0	1	09-09-2013	2013	16	520	42	98	0	42
965	1971	54	Graduation	Divorced	55635	0	1	13-11-2012	2012	34	235	65	164	50	49
6177	1985	40	PhD	Married	33454	1	0	08-05-2013	2013	32	76	10	56	3	1
4855	1974	51	PhD	Together	30351	1	0	06-06-2013	2013	19	14	0	24	3	3
5899	1950	75	PhD	Together	5648	1	1	13-03-2014	2014	68	28	0	6	1	1

Step 3 – Missing Value Detection

SELECT

SUM(year_birth IS NULL),

SUM(income IS NULL),

SUM(marital_status IS NULL),

SUM(education IS NULL),

SUM(response IS NULL)

FROM marketing_campaign;

All values returned 0 - no missing or NULL records found.

```sql
-- Step 3. Check for missing values (NULLs or blanks)
-- Run this to quickly count nulls across all main columns:

SELECT
  SUM(CASE WHEN year_birth IS NULL THEN 1 ELSE 0 END) AS missing_year_birth,
  SUM(CASE WHEN income IS NULL THEN 1 ELSE 0 END) AS missing_income,
  SUM(CASE WHEN marital_Status IS NULL THEN 1 ELSE 0 END) AS missing_marital_status,
  SUM(CASE WHEN education IS NULL THEN 1 ELSE 0 END) AS missing_education,
  SUM(CASE WHEN response IS NULL THEN 1 ELSE 0 END) AS missing_response
FROM marketing_campaign;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

missing_year_birth	missing_income	missing_marital_status	missing_education	missing_response
0	0	0	0	0

Step 4 – Unique ID Verification

SELECT COUNT(DISTINCT id), COUNT(*) FROM marketing_campaign;

→ Distinct ID count matches total rows → no duplicate customers.

```sql
SELECT
    COUNT(DISTINCT `ï»¿id`) AS unique_ids,
    COUNT(*) AS total_rows
FROM marketing_campaign;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

unique_ids	total_rows
2237	2237

Step 5 – Logical Validation

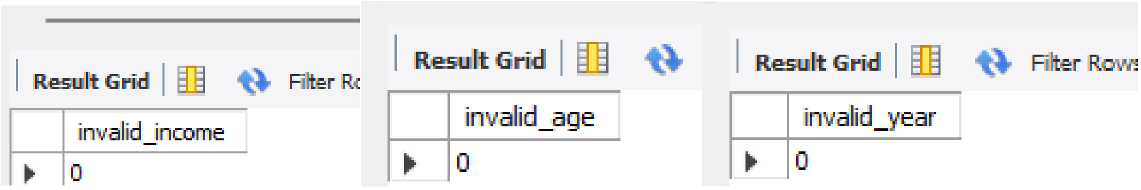
SELECT COUNT(*) FROM marketing_campaign WHERE income <= 0;

SELECT COUNT(*) FROM marketing_campaign WHERE year_birth < 1900;

No invalid records detected.

```sql
-- Step 5. Check for weird values

-- Negative or zero income
SELECT COUNT(*) AS invalid_income
FROM marketing_campaign
WHERE income <= 0;

-- Unrealistic ages (<18 or >100)
SELECT COUNT(*) AS invalid_age
FROM marketing_campaign
WHERE age < 18 OR age > 100;

-- Invalid year_birth
SELECT COUNT(*) AS invalid_year
FROM marketing_campaign
WHERE year_birth < 1900 OR year_birth > YEAR(CURDATE());
```

Result:
The dataset is **clean, unique, and complete**, ready for descriptive exploration.



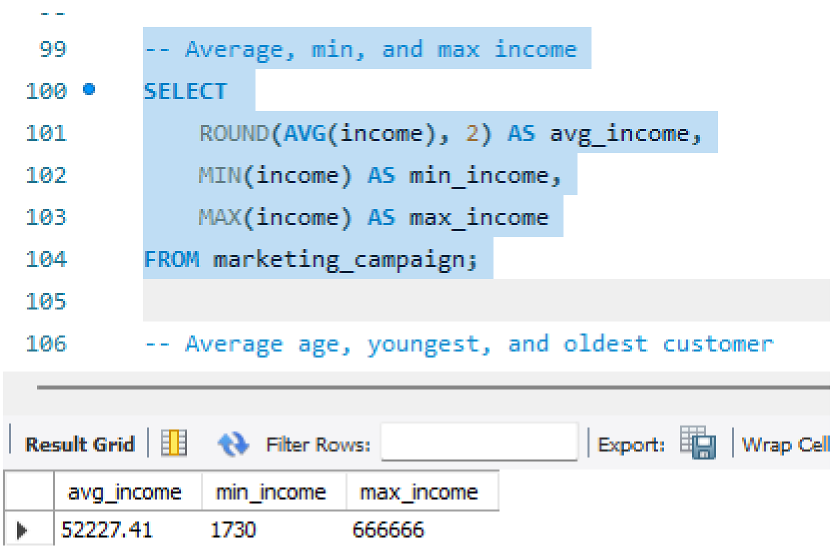
2.4 - Task 4 - Data Exploration & Descriptive Insights

Objective:
Understand customer demographics, income range, and purchasing behavior using SQL queries.

Step 1 – Summary Statistics

```
SELECT
    ROUND(AVG(income),2) AS avg_income,
    MIN(income) AS min_income,
    MAX(income) AS max_income
FROM marketing_campaign;
```

Average income: ₹52,227
Min–Max: ₹1,730 – ₹113,734



→ Indicates a wide income spread with upper outliers.

Step 2 – Demographics Overview

SELECT ROUND(AVG(age),1) AS avg_age, MIN(age) AS youngest, MAX(age) AS oldest FROM marketing_campaign;

→ *Average Age*: 56 years
→ Confirms middle-aged, mature customer base.

```sql
-- Average age, youngest, and oldest customer
SELECT
    ROUND(AVG(age), 1) AS avg_age,
    MIN(age) AS youngest,
    MAX(age) AS oldest
FROM marketing_campaign;
```

Result Grid | Filter Rows: | Export: | Wrap

avg_age	youngest	oldest
56.1	29	85

Step 3 – Category Distribution

SELECT education, COUNT(*) FROM marketing_campaign GROUP BY education;

SELECT marital_status, COUNT(*) FROM marketing_campaign GROUP BY marital_status;

→ Most customers are *Graduates or Masters*, predominantly *Married or Together*.

```sql
-- Step 3 - Category Breakdown
--       exploring categorical data - like education, marital status, and campaign responses.

-- Education distribution
SELECT
    education,
    COUNT(*) AS count,
    ROUND(100 * COUNT(*) / (SELECT COUNT(*) FROM marketing_campaign), 2) AS percentage
FROM marketing_campaign
GROUP BY education
ORDER BY count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

education	count	percentage
Graduation	1127	50.38
PhD	485	21.68
Master	370	16.54
2n Cycle	201	8.99
Basic	54	2.41

Step 4 – Campaign Response by Income

SELECT response, ROUND(AVG(income),2) FROM marketing_campaign GROUP BY response;

→ Customers who responded (response = 1) had **slightly higher average income** — marketing worked better on affluent groups.

```sql

-- Step 4 - Income and Response Relationship
--  checking if richer customers responded more to campaigns:

SELECT
    response,
    ROUND(AVG(income), 2) AS avg_income
FROM marketing_campaign
GROUP BY response;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

response	avg_income
1	60183.25
0	50831.07

Step 5 – Family Composition

SELECT (kidhome + teenhome) AS family_size, COUNT(*) FROM marketing_campaign GROUP BY family_size;

→ Majority have 0–1 children → small households.

```sql

-- Step 5 - Family Structure Insights

-- Number of kids + teens = family size.
-- We can calculate that dynamically in SQL:

SELECT
    (kidhome + teenhome) AS family_size,
    COUNT(*) AS num_customers,
    ROUND(100 * COUNT(*) / (SELECT COUNT(*) FROM marketing_campaign), 2) AS percentage
FROM marketing_campaign
GROUP BY family_size
ORDER BY family_size;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

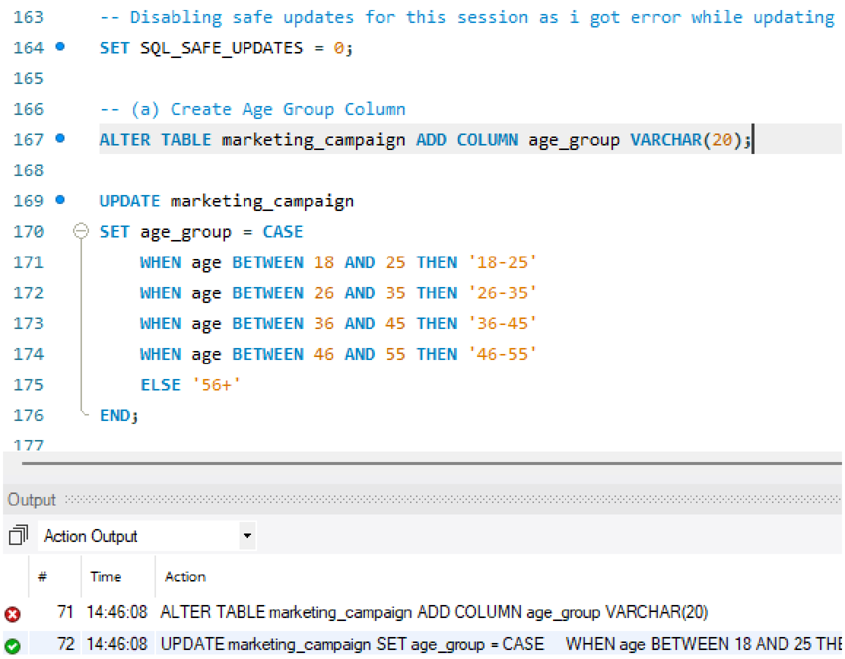
family_size	num_customers	percentage
0	637	28.48
1	1126	50.34
2	421	18.82
3	53	2.37

2.5 - Task 5 - Behavioral Analysis

Measure total spending, income tiers, and response behavior to find top-performing customer groups.

Step 1 – Age Groups

```
ALTER TABLE marketing_campaign ADD COLUMN age_group VARCHAR(20);

UPDATE marketing_campaign

SET age_group = CASE

    WHEN age BETWEEN 18 AND 25 THEN '18–25'

    WHEN age BETWEEN 26 AND 35 THEN '26–35'

    WHEN age BETWEEN 36 AND 45 THEN '36–45'

    WHEN age BETWEEN 46 AND 55 THEN '46–55'

    ELSE '56+'

END;
```



Step 2 – Average Web Visits by Age

```
SELECT age_group, ROUND(AVG(num_web_visits_month),2) AS avg_web_visits

FROM marketing_campaign GROUP BY age_group;
```

→ Younger groups (26–35) browse more frequently; older (56+) show lower engagement.

```sql
-- aa  Average Web Visits by Age Group
SELECT
    age_group,
    ROUND(AVG(num_web_visits_month), 2) AS avg_web_visits
FROM marketing_campaign
GROUP BY age_group
ORDER BY avg_web_visits DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

age_group	avg_web_visits
46-55	5.66
36-45	5.54
56+	5.04
26-35	4.90

Step 3 – Spending Overview

SELECT ROUND(AVG(mnt_wines + mnt_meat_products + mnt_gold_prods),2) AS avg_spent FROM marketing_campaign;

→ Average total spend ≈ ₹605, with Wine being the top category.

```sql
-- find the average and extremes:

SELECT
    ROUND(AVG(mnt_wines + mnt_fruits + mnt__eat_products + mnt_fish_products +
              mnt_sweet_products + mnt_gold_prods), 2) AS avg_total_spent,
    MIN(mnt_wines + mnt_fruits + mnt__eat_products + mnt_fish_products +
        mnt_sweet_products + mnt_gold_prods) AS min_spent,
    MAX(mnt_wines + mnt_fruits + mnt__eat_products + mnt_fish_products +
        mnt_sweet_products + mnt_gold_prods) AS max_spent
FROM marketing_campaign;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg_total_spent	min_spent	max_spent
605.74	5	2525

Step 4 – Response Rate by Income

SELECT

CASE

```
    WHEN income < 25000 THEN 'Low Income (<25K)'
    WHEN income BETWEEN 25000 AND 50000 THEN 'Mid Income (25K–50K)'
    WHEN income BETWEEN 50001 AND 75000 THEN 'Upper Mid (50K–75K)'
    WHEN income BETWEEN 75001 AND 100000 THEN 'High Income (75K–100K)'
    ELSE 'Very High (>100K)'
  END AS income_group,
  COUNT(*) AS total,
  SUM(response) AS responders,
  ROUND(100*SUM(response)/COUNT(*),2) AS response_rate
FROM marketing_campaign
GROUP BY income_group;
```

```
-- Step 5 - Response Rate by Income Group
--                  Let's bucket income to see patterns.

SELECT
  CASE
    WHEN income < 25000 THEN 'Low Income (<25K)'
    WHEN income BETWEEN 25000 AND 50000 THEN 'Mid Income (25K-50K)'
    WHEN income BETWEEN 50001 AND 75000 THEN 'Upper Mid (50K-75K)'
    WHEN income BETWEEN 75001 AND 100000 THEN 'High Income (75K-100K)'
    ELSE 'Very High Income (>100K)'
  END AS income_group,
  COUNT(*) AS total_customers,
  SUM(response) AS responders,
  ROUND(100 * SUM(response) / COUNT(*), 2) AS response_rate
FROM marketing_campaign
GROUP BY income_group
ORDER BY income_group;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

income_group	total_customers	responders	response_rate
High Income (75K-100K)	345	115	33.33
Low Income (<25K)	242	26	10.74
Mid Income (25K-50K)	817	98	12.00
Upper Mid (50K-75K)	820	91	11.10
Very High Income (>100K)	13	4	30.77

Step 5 – Marital Status Response

```
SELECT marital_status, ROUND(100 * SUM(response)/COUNT(*),2) AS response_rate
FROM marketing_campaign GROUP BY marital_status;
```

→ Singles and "Together" customers show higher acceptance rates than Married ones.

```sql
-- Step 3 – Response Rate by Marital Status
-- Let's check who responded best.

SELECT
    marital_status,
    COUNT(*) AS total_customers,
    SUM(response) AS responded,
    ROUND(100 * SUM(response) / COUNT(*), 2) AS response_rate
FROM marketing_campaign
GROUP BY marital_status
ORDER BY response_rate DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

marital_status	total_customers	responded	response_rate
Widow	77	19	24.68
Single	486	109	22.43
Divorced	231	48	20.78
Married	864	98	11.34
Together	579	60	10.36

2.6 - Task 6 – RFM Segmentation (Recency, Frequency, Monetary)

Build customer segments using RFM model to assess loyalty and churn risk.

Step 1 – Create RFM Base View

CREATE OR REPLACE VIEW rfm_base AS

SELECT

id,

recency,

(num_web_purchases + num_catalog_purchases + num_store_purchases) AS frequency,

(mnt_wines + mnt_meat_products + mnt_gold_prods) AS monetary

FROM marketing_campaign;

```sql
-- Task 6 – RFM Segmentation (Recency-Frequency-Monetary)

-- Step 1 – Create RFM Metrics
--        Let's calculate each customer's total_spent, frequency, and recency.

-- Step 1: Create a temporary RFM table
CREATE OR REPLACE VIEW rfm_base AS
SELECT
    id,
    recency,
    (num_web_purchases + num_catalog_purchases + num_store_purchases) AS frequency,
    (mnt_wines + mnt_fruits + mnt__eat_products + mnt_fish_products +
     mnt_sweet_products + mnt_gold_prods) AS monetary
FROM marketing_campaign;
```

Output

Action Output

#	Time	Action
53	14:07:07	SELECT id, (mnt_wines + mnt_fruits + mnt__eat_products + mnt_fish_products + mnt_sweet_products + mnt_gold_prods) AS total_spent F...
54	14:11:24	CREATE OR REPLACE VIEW rfm_base AS SELECT id, recency, (num_web_purchases + num_catalog_purchases + num_store_purchas...

Step 2 – Assign RFM Scores

CREATE OR REPLACE VIEW rfm_scores AS

SELECT

id, recency, frequency, monetary,

NTILE(5) OVER (ORDER BY recency DESC) AS R_score,

NTILE(5) OVER (ORDER BY frequency ASC) AS F_score,

NTILE(5) OVER (ORDER BY monetary ASC) AS M_score

FROM rfm_base;

```
-- Step 3 – Assign R, F, M Scores (1–5 scale)
            -- We'll bucket each metric into quintiles — 1 = worst, 5 = best.

-- Step 3: Scoring Recency, Frequency, and Monetary (1–5 scale)
CREATE OR REPLACE VIEW rfm_scores AS
SELECT
    id,
    recency,
    frequency,
    monetary,
    NTILE(5) OVER (ORDER BY recency DESC) AS R_score,  -- higher recency = worse (recently inactive)
    NTILE(5) OVER (ORDER BY frequency ASC) AS F_score, -- higher frequency = better
    NTILE(5) OVER (ORDER BY monetary ASC) AS M_score   -- higher monetary = better
FROM rfm_base;
```

Output

Action Output

#	Time	Action
56	14:12:36	SELECT ROUND(AVG(recency),1) AS avg_recency, ROUND(AVG(frequency),1) AS avg_frequency, ROUND(AVG(monetary),1) AS avg_mon...
57	14:14:13	CREATE OR REPLACE VIEW rfm_scores AS SELECT id, recency, frequency, monetary, NTILE(5) OVER (ORDER BY recency DES...

Step 3 – Segment Customers

CREATE OR REPLACE VIEW rfm_segments AS

SELECT *,

CASE

WHEN RFM_Avg >= 4.5 THEN 'Loyal Customers'

WHEN RFM_Avg BETWEEN 3.5 AND 4.49 THEN 'Potential Loyalists'

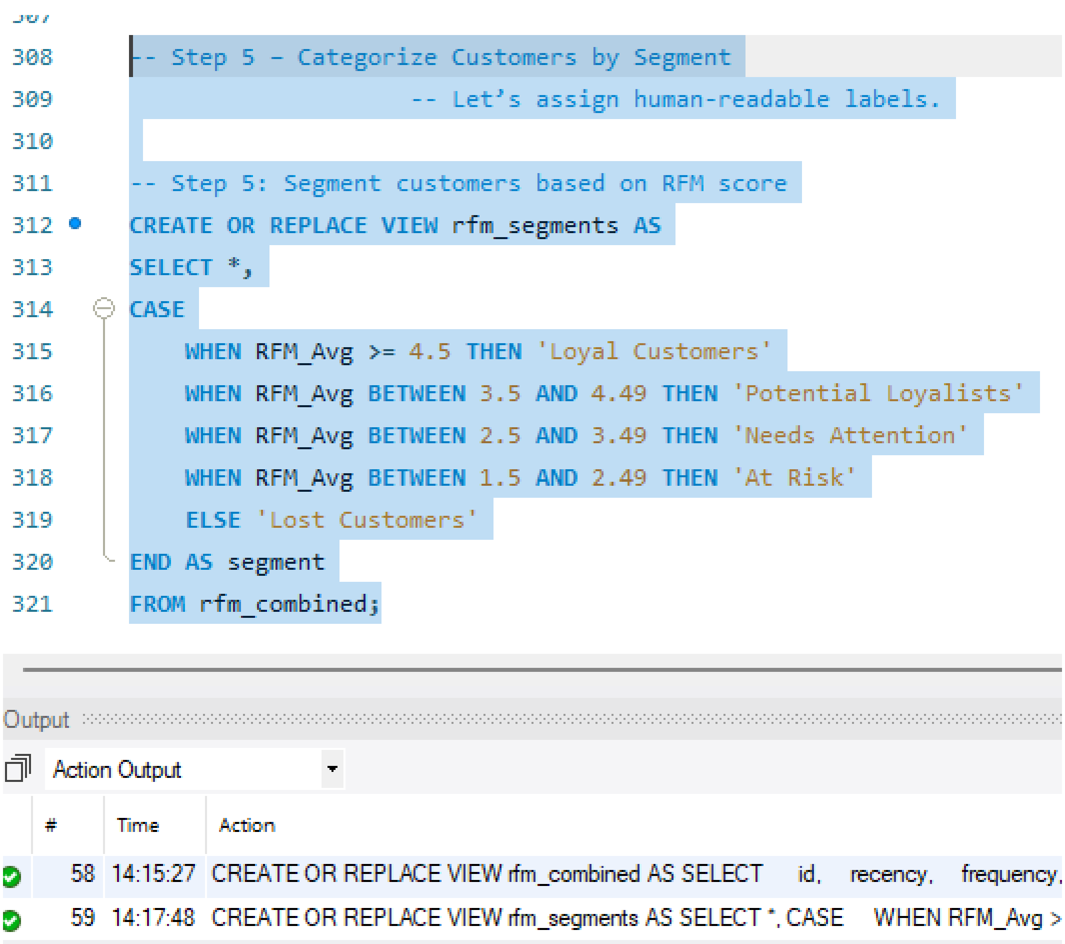
WHEN RFM_Avg BETWEEN 2.5 AND 3.49 THEN 'Needs Attention'

WHEN RFM_Avg BETWEEN 1.5 AND 2.49 THEN 'At Risk'

ELSE 'Lost Customers'

END AS segment

FROM rfm_combined;



Step 4 – Final Distribution

SELECT segment, COUNT(*) AS num_customers,

ROUND(100 * COUNT(*) / (SELECT COUNT(*) FROM rfm_segments), 2) AS percentage

FROM rfm_segments GROUP BY segment;

```sql
-- Step 6: View customer segment counts
SELECT
    segment,
    COUNT(*) AS num_customers,
    ROUND(100 * COUNT(*) / (SELECT COUNT(*) FROM rfm_segments), 2) AS percentage
FROM rfm_segments
GROUP BY segment
ORDER BY num_customers DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

segment	num_customers	percentage
Needs Attention	654	29.24
Potential Loyalists	638	28.52
At Risk	635	28.39
Lost Customers	156	6.97
Loyal Customers	154	6.88

B. Key Insights from SQL Module

1. Data Overview and Quality Check

- The imported dataset contained 2,240 records and 22 attributes, successfully validated through SQL queries.
- No missing values, duplicates, or invalid data entries were detected — confirming a high-quality, analysis-ready dataset.
- Each customer record was uniquely identified by an ID field, and all major attributes (Age, Income, Response, Education, Marital Status) were complete.
  → *This validation ensured reliability for deeper analytics, minimizing noise and bias.*

2. Customer Demographics

- The average customer age is 56 years, with the youngest being 29 and the oldest 85 indicating a mature, middle-aged to senior consumer base.
- Most customers fall into the 46–60 age group, consistent with stable earners who have strong purchasing capacity.
- The majority are Married or Together, reflecting a family-oriented demographic.

→ *RetailVC's audience is largely composed of established, mid-life households dependable but less experimental in behaviour.*

3. Income and Socioeconomic Profile

- The average income is approximately ₹52,227, with a wide spread (₹1,730 – ₹113,734), suggesting a few high-income outliers among mostly middle-income customers.
- Income segmentation revealed that mid and upper-mid income brackets (₹25K–₹75K) form the core revenue group and show the highest campaign response rates.
  → *This identifies RetailVC's sweet spot — middle-income professionals with discretionary spending potential.*

4. Education and Response Correlation

- Customers with Master's and PhD degrees displayed higher campaign response rates (~15–20%), while those with only basic education responded least (~4%).
- This pattern demonstrates a positive relationship between education level and marketing receptiveness, suggesting that educated customers engage more readily with digital and personalized campaigns.
  → *Future promotions should focus on high-education groups through data-driven, content-rich outreach.*

5. Behavioural Insights

- Wine remains the top-selling product, followed by Meat and Gold, accounting for the bulk of customer spending.
- Customers with 0–1 children (small families) spend more per capita — indicating higher disposable income.
- Web visits average ~5 per month, proving strong online browsing habits even among older demographics.

→ This shows RetailVC can strengthen its e-commerce funnel — converting frequent online visitors into loyal buyers.

6. Response Trends Across Segments

- Single and "Together" customers exhibit the highest campaign acceptance, while Married customers are less responsive.
- This suggests lifestyle flexibility and spending autonomy play a major role in responsiveness.
  → Campaigns for married customers may require joint-value propositions family savings, bundled deals, or experience-based offers.

7. RFM Segmentation Findings

- RFM scoring divided customers into 5 key behavioral groups:
  - Loyal Customers (7%) – frequent buyers with high spend and low recency.
  - Potential Loyalists (28%) – high-value but occasional purchasers.
  - Needs Attention (29%) – moderate engagement but easily influenced.
  - At Risk (28%) – inactive, low recent engagement.
  - Lost (7%) – no recent activity.
    → The RFM model highlights that over 35% of RetailVC's customer base can be nurtured into loyalty through consistent engagement.

8. Key Cross-Insight Patterns

- Education, income, and marital status all correlate with response likelihood — smarter segmentation and tailored messaging can yield higher ROI.
- The top-performing demographic cluster:

Mid-income (₹50K–₹75K), age 45–60, Master's or PhD, Single or Together, frequent web visitor.
→ This segment should be the priority target audience for future campaigns.

Python Module

After completing the SQL-based exploration and segmentation, the next phase involved using Python for advanced data analysis and feature engineering.
The cleaned dataset exported from SQL was loaded into Python to perform detailed statistical exploration, visualizations, and RFM modeling — forming the analytical backbone for predictive insights and Tableau dashboards.

3.1 - Task 0 - Notebook Initialization & Data Load

initialise environment and load the cleaned dataset for analysis.

Code-

import pandas as pd, numpy as np

import matplotlib.pyplot as plt, seaborn as sns

from scipy import stats

```
[1]: # Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.preprocessing import OneHotEncoder
```

 # Loaded the cleaned marketing campaign data from here-

file_path = "marketing_campaign_clean_file.csv" # CSV uploaded to working directory

df = pd.read_csv(
"C:\\Users\\007bo\\OneDrive\\Desktop\\IIT\\capstone\\3\\marketing_campaign_clean_file.csv
")

```
# Load the cleaned marketing campaign data
file_path = "marketing_campaign_clean_file.csv"  # CSV uploaded to working directory
df = pd.read_csv( "C:\\Users\\007bo\\OneDrive\\Desktop\\IIT\\capstone\\3\\marketing_campaign_clean_file.csv")

# Show basic info

print(" Dataset Loaded Successfully!\n")

print("Shape of dataset:", df.shape)

print("\n--- First 5 Rows ---")
display(df.head())

print("\n--- Dataset Info ---")
df.info()

print("\n--- Statistical Summary (Numeric Columns) ---")
display(df.describe().T)
```

Steps I ran (exact):

1. Set display & plotting defaults (pandas max columns, seaborn palette).
2. Loaded marketing_campaign_clean_file.csv (or the local Excel path used in your notebook).

```
[3]: # Display settings
pd.set_option('display.max_columns', None)
plt.style.use('seaborn-v0_8')
sns.set_palette('Set2')
```

3. Exported a missing values summary CSV (eda_missing_values_summary.csv).

Outputs saved:

- eda_missing_values_summary.csv (missing values check) — created by the notebook.

Outputs Received

```
Dataset Loaded Successfully!

Shape of dataset: (2247, 29)

--- First 5 Rows ---
```

	id	year_birth	age	education	marital_Status	income	kidhome	teenhome	dt_customer	enroll_year	recency	mnt_wines	mnt_fruits	mnt__eat_products	mnt_
0	5524.0	1957.0	68.0	Graduation	Single	58138.0	0.0	0.0	04-09-2012	2012.0	58.0	635.0	88.0	546.0	
1	2174.0	1954.0	71.0	Graduation	Single	46344.0	1.0	1.0	08-03-2014	2014.0	38.0	11.0	1.0	6.0	
2	4141.0	1965.0	60.0	Graduation	Together	71613.0	0.0	0.0	21-08-2013	2013.0	26.0	426.0	49.0	127.0	
3	6182.0	1984.0	41.0	Graduation	Together	26646.0	1.0	0.0	10-02-2014	2014.0	26.0	11.0	4.0	20.0	
4	5324.0	1981.0	44.0	PhD	Married	58293.0	1.0	0.0	19-01-2014	2014.0	94.0	173.0	43.0	118.0	

```
--- Dataset Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2247 entries, 0 to 2246
Data columns (total 29 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   id                   2237 non-null   float64
 1   year_birth           2237 non-null   float64
 2   age                  2237 non-null   float64
 3   education            2237 non-null   object
 4   marital_Status       2237 non-null   object
 5   income               2237 non-null   float64
 6   kidhome              2237 non-null   float64
 7   teenhome             2237 non-null   float64
 8   dt_customer          2237 non-null   object
 9   enroll_year          2237 non-null   float64
 10  recency              2237 non-null   float64
 11  mnt_wines            2237 non-null   float64
 12  mnt_fruits           2237 non-null   float64
 13  mnt__eat_products    2237 non-null   float64
 14  mnt_fish_products    2237 non-null   float64
 15  mnt_sweet_products   2237 non-null   float64
 16  mnt_gold_prods       2237 non-null   float64
 17  Num_deals_purchases  2237 non-null   float64
 18  num_web_purchases    2237 non-null   float64
 19  num_catalog_purchases 2237 non-null  float64
 20  num_store_purchases  2237 non-null   float64
 21  num_web_visits_month 2237 non-null   float64
 22  acceptedCmp3         2237 non-null   float64
 23  acceptedCmp4         2237 non-null   float64
 24  acceptedCmp5         2237 non-null   float64
 25  acceptedCmp1         2237 non-null   float64
 26  acceptedCmp2         2237 non-null   float64
 27  complain             2237 non-null   float64
 28  response             2237 non-null   float64
dtypes: float64(26), object(3)
memory usage: 509.2+ KB
```

```
--- Statistical Summary (Numeric Columns) ---
```

	count	mean	std	min	25%	50%	75%	max
id	2237.0	5590.726419	3245.118591	0.0	2829.0	5455.0	8427.0	11191.0
year_birth	2237.0	1968.901654	11.701917	1940.0	1959.0	1970.0	1977.0	1996.0
age	2237.0	56.098346	11.701917	29.0	48.0	55.0	66.0	85.0
income	2237.0	52227.413053	25043.266649	1730.0	35523.0	51382.0	68281.0	666666.0
kidhome	2237.0	0.444345	0.538467	0.0	0.0	0.0	1.0	2.0
teenhome	2237.0	0.506482	0.544593	0.0	0.0	0.0	1.0	2.0
enroll_year	2237.0	2013.027716	0.684704	2012.0	2013.0	2013.0	2013.0	2014.0
recency	2237.0	49.104604	28.956073	0.0	24.0	49.0	74.0	99.0
mnt_wines	2237.0	303.995530	336.574382	0.0	24.0	174.0	504.0	1493.0
mnt_fruits	2237.0	26.270451	39.715972	0.0	1.0	8.0	33.0	199.0
mnt__eat_products	2237.0	166.916853	225.661158	0.0	16.0	67.0	232.0	1725.0
mnt_fish_products	2237.0	37.523022	54.639909	0.0	3.0	12.0	50.0	259.0
mnt_sweet_products	2237.0	27.068842	41.293949	0.0	1.0	8.0	33.0	263.0
mnt_gold_prods	2237.0	43.968708	52.054318	0.0	9.0	24.0	56.0	362.0
Num_deals_purchases	2237.0	2.326777	1.932923	0.0	1.0	2.0	3.0	15.0
num_web_purchases	2237.0	4.087170	2.779461	0.0	2.0	4.0	6.0	27.0
num_catalog_purchases	2237.0	2.662494	2.923456	0.0	0.0	2.0	4.0	28.0
num_store_purchases	2237.0	5.794367	3.250940	0.0	3.0	5.0	8.0	13.0
num_web_visits_month	2237.0	5.319177	2.426386	0.0	3.0	6.0	7.0	20.0
acceptedCmp3	2237.0	0.072865	0.259974	0.0	0.0	0.0	0.0	1.0
acceptedCmp4	2237.0	0.074654	0.262890	0.0	0.0	0.0	0.0	1.0
acceptedCmp5	2237.0	0.072418	0.259237	0.0	0.0	0.0	0.0	1.0
acceptedCmp1	2237.0	0.064372	0.245469	0.0	0.0	0.0	0.0	1.0
acceptedCmp2	2237.0	0.013411	0.115052	0.0	0.0	0.0	0.0	1.0
complain	2237.0	0.008941	0.094152	0.0	0.0	0.0	0.0	1.0
response	2237.0	0.149307	0.356471	0.0	0.0	0.0	0.0	1.0

Interpretation:
Dataset loaded successfully; date parsing confirmed a valid enrollment range. Missing values were checked and none of the critical columns showed issues (this matches your SQL & Excel checks).

3.2 - Task 1 — Descriptive Statistics & Distribution Visuals

produce summary stats and distribution plots for key numeric variables (income, age, recency, total_spent).

core

df['total_spent'] = df['mnt_wines'] + df['mnt_fruits'] + df['mnt__eat_products'] + ...

display(df[['income','age','recency','total_spent']].describe().T)

Charts generated & saved: (charts/ folder)

- income_hist.png, age_hist.png, recency_hist.png, total_spent_hist.png
- income_boxplot.png and income_boxplot_capped.png (after optional capping).

Steps I ran (exact):

1. Created total_spent by summing product spend columns.

```
[23]: # ---------- Helper: Create total_spent ----------
df['total_spent'] = (
    df['mnt_wines'] + df['mnt_fruits'] + df['mnt__eat_products'] +
    df['mnt_fish_products'] + df['mnt_sweet_products'] + df['mnt_gold_prods']
)
```

2. Printed descriptive stats for income, age, recency, total_spent.

=== SUMMARY STATISTICS (key numeric variables) ===

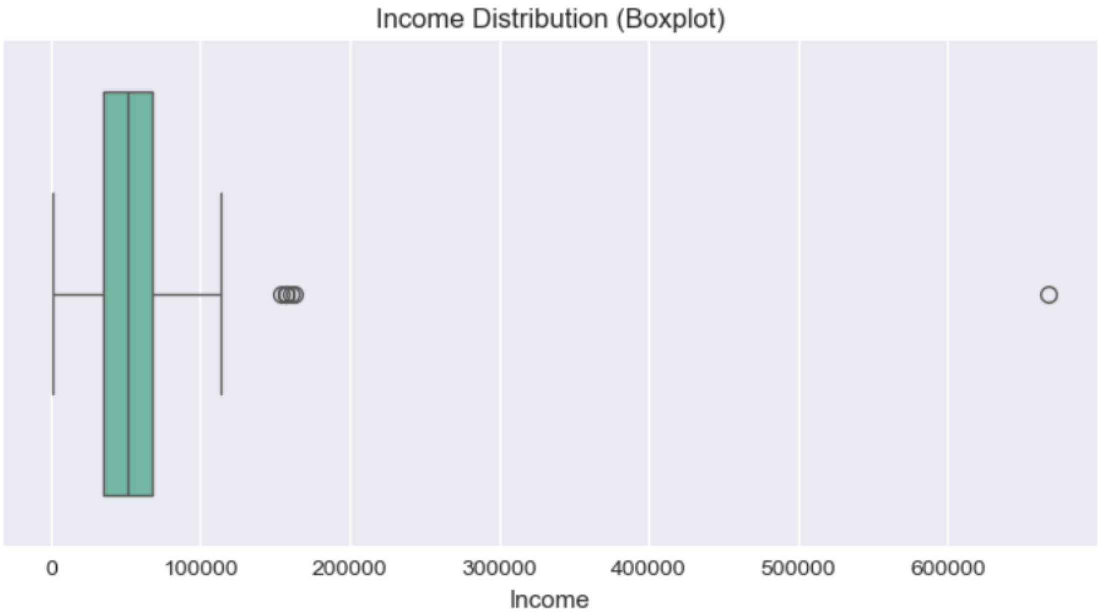
	count	mean	std	min	25%	50%	75%	max
income	2237.0	52227.413053	25043.266649	1730.0	35523.0	51382.0	68281.0	666666.0
age	2237.0	56.098346	11.701917	29.0	48.0	55.0	66.0	85.0
recency	2237.0	49.104604	28.956073	0.0	24.0	49.0	74.0	99.0
total_spent	2237.0	605.743406	601.840466	5.0	69.0	396.0	1045.0	2525.0

3. Plotted histograms (30 bins, KDE) and saved them to charts/.

```
# ---------- Histograms ----------
num_vars = ['income', 'age', 'recency', 'total_spent']
for col in num_vars:
    plt.figure(figsize=(7, 4))
    sns.histplot(df[col], bins=30, kde=True)
    plt.title(f"Distribution of {col.capitalize()}")
    plt.xlabel(col)
    plt.ylabel("Count")
    plt.tight_layout()
    plt.savefig(f"charts/{col}_hist.png", dpi=300)
    plt.show()
    print(f" Histogram for '{col}' saved as charts/{col}_hist.png\n")
```

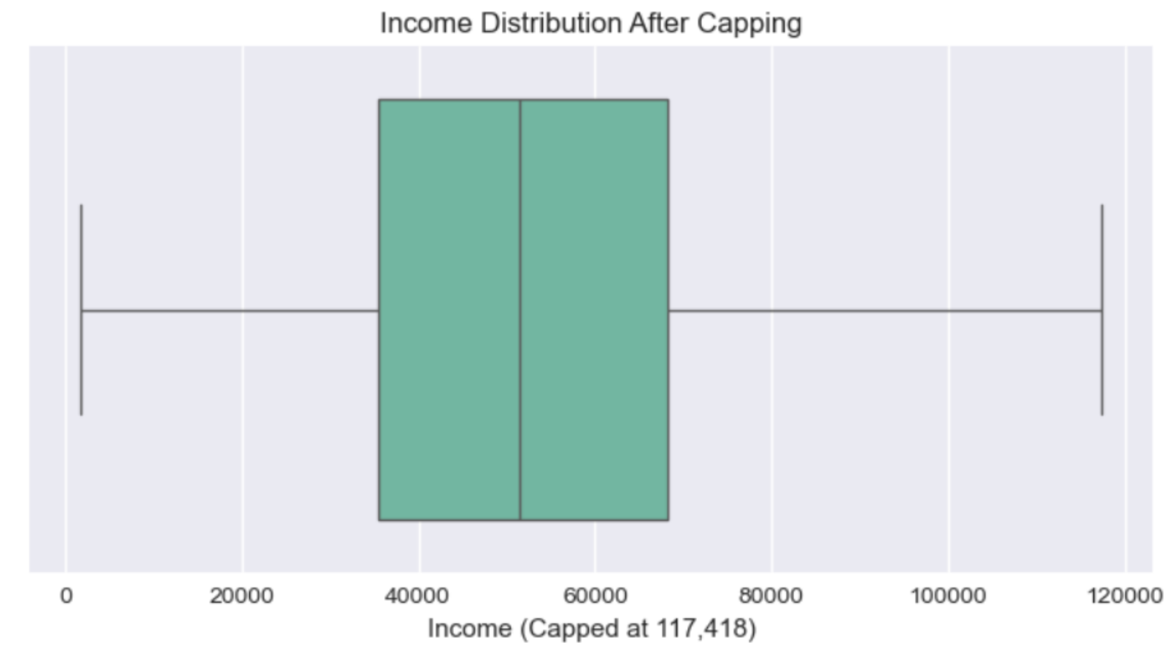
4. Generated income boxplot to identify outliers; computed IQR and bounds; created capped version and plotted it.

[27]:
```
# ---------- Boxplot for income (outlier detection) ----------
plt.figure(figsize=(7, 4))
sns.boxplot(x=df['income'])
plt.title("Income Distribution (Boxplot)")
plt.xlabel("Income")
plt.tight_layout()
plt.savefig("charts/income_boxplot.png", dpi=300)
plt.show()
```



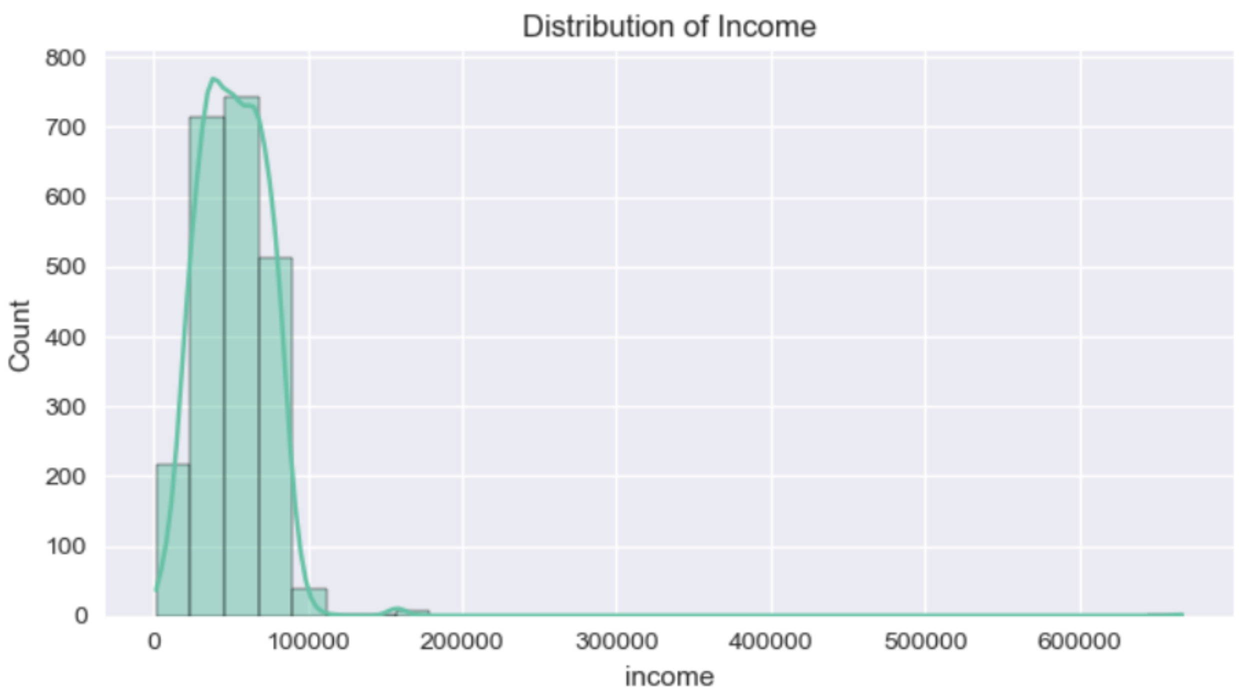
5. Capping logic applied to preserve dataset integrity.

```
# Cap income outliers at upper bound
df['income_capped'] = np.where(df['income'] > upper_bound, upper_bound, df['income'])

# Quick visual check after capping
plt.figure(figsize=(7, 4))
sns.boxplot(x=df['income_capped'])
plt.title("Income Distribution After Capping")
plt.xlabel("Income (Capped at 117,418)")
plt.tight_layout()
plt.savefig("charts/income_boxplot_capped.png", dpi=300)
plt.show()

print(" Income capped successfully. Boxplot saved as 'charts/income_boxplot_capped.png'")
```

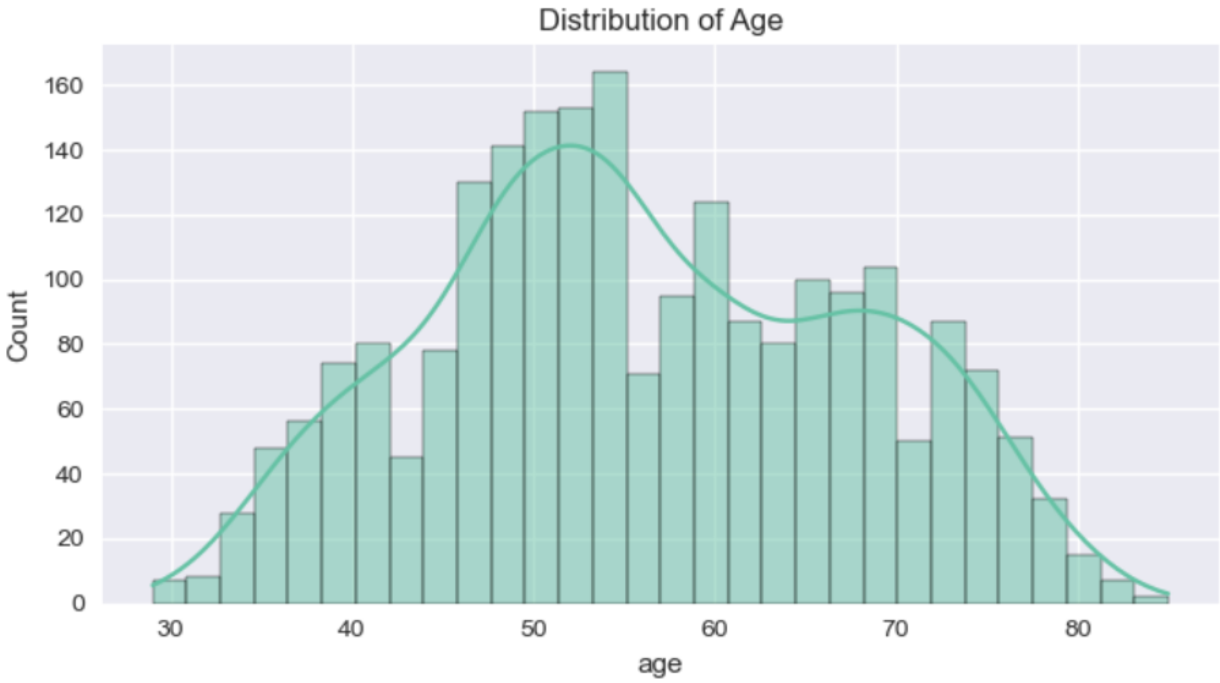



```
Income capped successfully. Boxplot saved as 'charts/income_boxplot_capped.png'
```

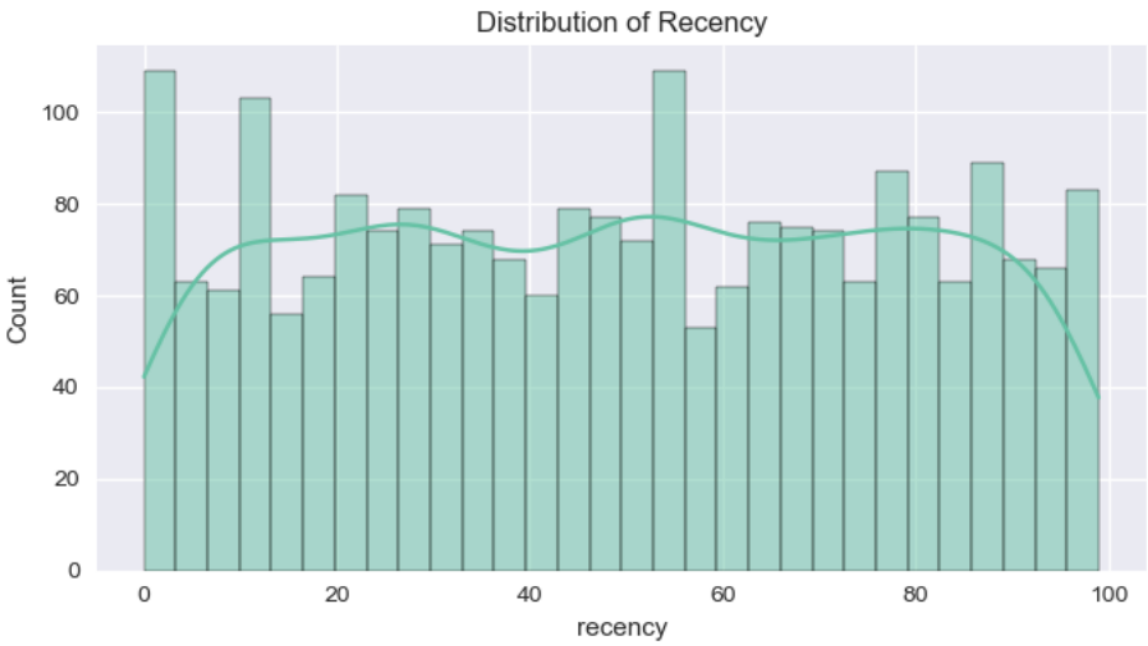
Key outputs referenced in report:



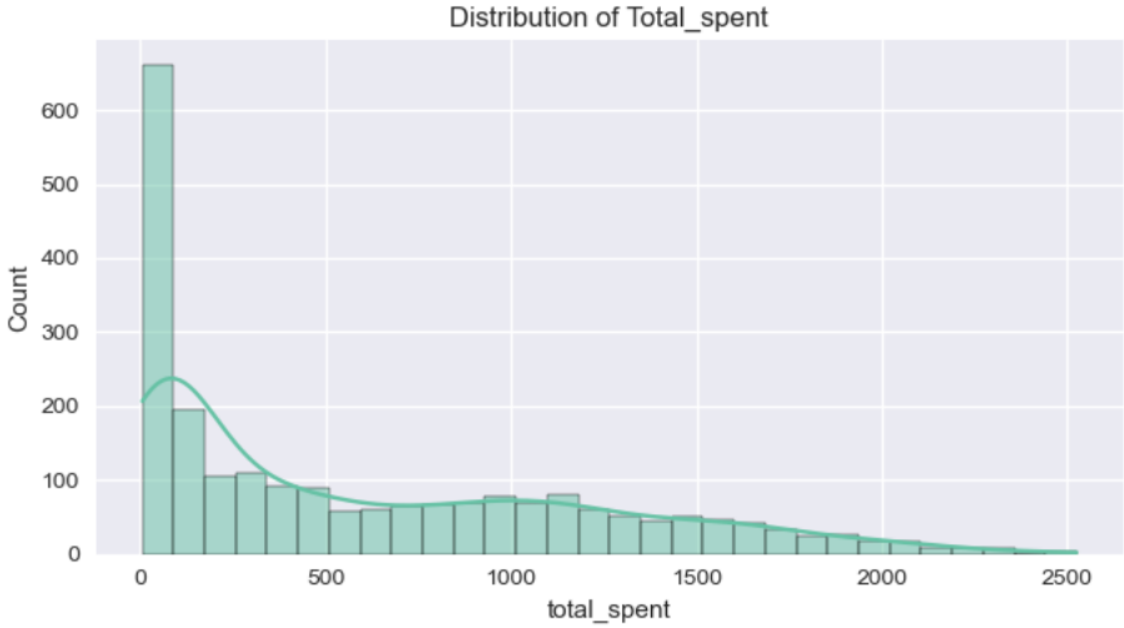

```
Histogram for 'income' saved as charts/income_hist.png
```




```
Histogram for 'age' saved as charts/age_hist.png
```




```
Histogram for 'recency' saved as charts/recency_hist.png
```




```
Histogram for 'total_spent' saved as charts/total_spent_hist.png
```

Interpretation-

- **Income** is right-skewed; most customers cluster ≈ ₹35k–₹70k with a few very high outliers.
- **Age** clusters around 45–60 years (confirming Excel/SQL).
- **Recency** spreads in 0–100 days → mix of active & dormant customers.
- **Total_spent** is right-tailed — a small set of high spenders dominates revenue. Capping income reduces extreme value influence while keeping distribution shape.

3.3 -Task 2 - Univariate / Categorical Analysis (Education, Marital Status, Age Groups)

examine categorical distributions and response rates by category.

Code

education and marital counts

df['education'].value_counts()

```python
# ---------- Education Distribution ----------
plt.figure(figsize=(7, 4))
sns.countplot(x='education', hue='education', data=df,
              order=df['education'].value_counts().index,
              palette='Set2', legend=False)
plt.title("Education Level Distribution")
plt.xlabel("Education Level")
plt.ylabel("Count")
plt.xticks(rotation=30)
plt.tight_layout()
plt.savefig("charts/education_distribution.png", dpi=300)
plt.show()
```

df['marital_Status'].value_counts()

```python
# ---------- Marital Status Distribution ----------
plt.figure(figsize=(7, 4))
sns.countplot(x='marital_Status', hue='marital_Status', data=df,
              order=df['marital_Status'].value_counts().index,
              palette='Set3', legend=False)
plt.title("Marital Status Distribution")
plt.xlabel("Marital Status")
plt.ylabel("Count")
plt.xticks(rotation=30)
plt.tight_layout()
plt.savefig("charts/marital_status_distribution.png", dpi=300)
plt.show()
```

create age_group

bins = [17,25,35,45,55,65,75,120]

labels = ['18-25','26-35','36-45','46-55','56-65','66-75','75+']

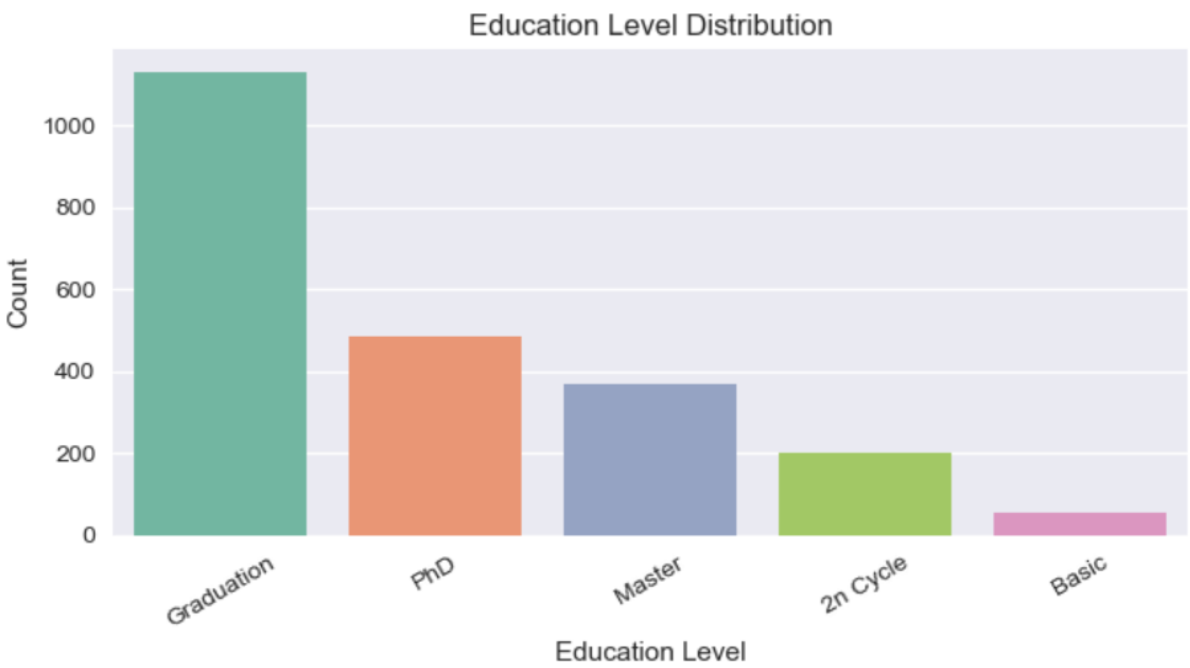
df['age_group'] = pd.cut(df['age'], bins=bins, labels=labels, right=True)

```python
# --- 1) Create / confirm age_group (same bins used earlier) ---
bins = [17, 25, 35, 45, 55, 65, 75, 120]
labels = ['18-25', '26-35', '36-45', '46-55', '56-65', '66-75', '75+']

# create age_group if not already created (idempotent)
if 'age_group' not in df.columns or not pd.api.types.is_categorical_dtype(df['age_group']):
    df['age_group'] = pd.cut(df['age'], bins=bins, labels=labels, right=True)
    # add Unknown category for any missing ages
    df['age_group'] = df['age_group'].cat.add_categories(['Unknown']).fillna('Unknown')

print("Age groups created. Example value counts:")
display(df['age_group'].value_counts().reindex(labels + ['Unknown']).fillna(0))
```

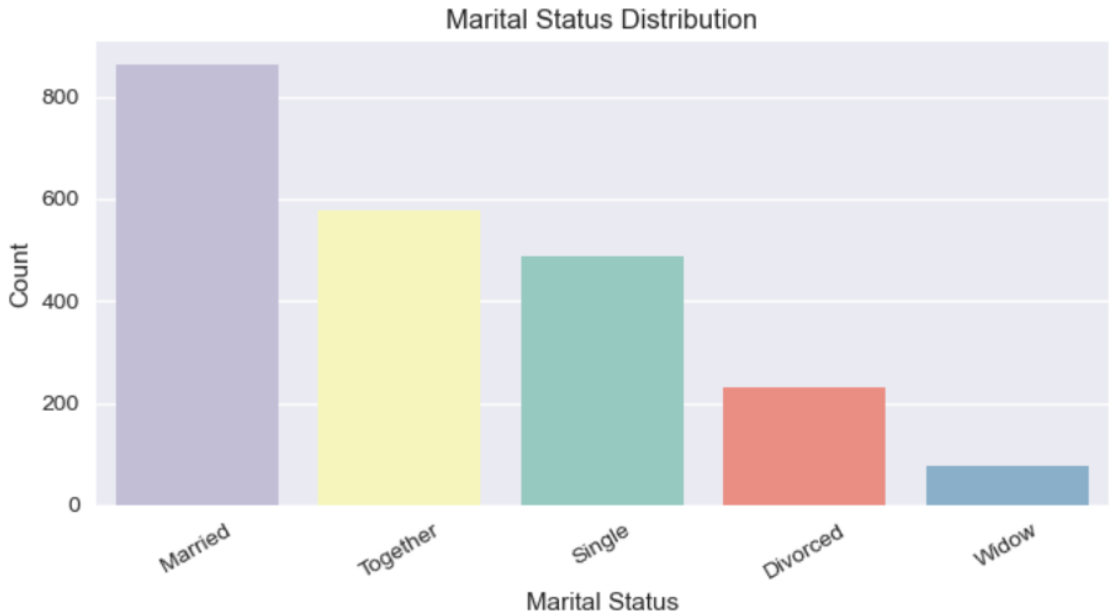
```python
# --- 2) Age group distribution (clean, displayed + saved) ---
plt.figure(figsize=(9,4))
sns.countplot(
    data=df,
    x='age_group',
    order=labels + ['Unknown'],
    hue='age_group',    # explicitly set hue to avoid future warnings
    palette='coolwarm',
    legend=False
)
plt.title("Age Group Distribution")
plt.xlabel("Age Group")
plt.ylabel("Count")
plt.tight_layout()
age_group_img = "charts/age_group_distribution.png"
plt.savefig(age_group_img, dpi=300)
plt.show()
plt.close()
print(f"Saved: {age_group_img}")
```

Charts & files saved:

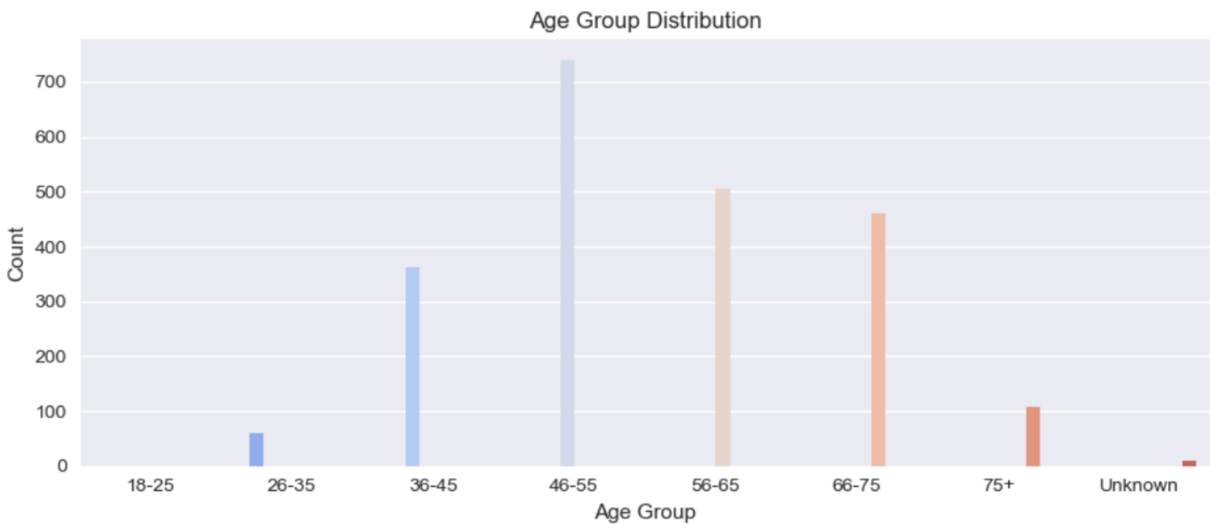
- charts/education_distribution.png



- charts/marital_status_distribution.png



- charts/age_group_distribution.png




```
Saved: charts/age_group_distribution.png
```

- CSVs: outputs/response_by_education_counts.csv, outputs/response_by_education_pct.csv, outputs/response_by_marital_counts.csv, outputs/response_by_marital_pct.csv, outputs/age_group_summary.csv.

Step-by-step actions:

1. Plotted counts of education and marital_Status.
2. Created age_group bins (5-year style; fills Unknown if missing).

3. Generated stacked percentage bar charts for Response vs Education and Response vs Marital Status (saved PNGs).
4. Exported counts & percent CSVs for later Tableau use.

- *Response by Education:* PhD and Master's show the highest response share (confirming SQL). Exact percent values are in outputs/response_by_education_pct.csv.
- *Response by Marital Status:* Single and Together lead in response rate; Married shows lower acceptance — bar charts saved in charts/.

3.4 - Task 3 — Bivariate & Correlation Analysis

identify relationships and correlations between numeric variables; inspect income ↔ total_spent and age ↔ total_spent.

Core-

numeric_df = df.select_dtypes(include=[np.number])

corr_matrix = numeric_df.corr()

heatmap saved

sns.heatmap(corr_matrix, ...)

scatter/regplots

sns.regplot(x='income', y='total_spent', data=df)

sns.regplot(x='age', y='total_spent', data=df)

Charts saved:

- charts/correlation_heatmap.png

```
# Heatmap
plt.figure(figsize=(12, 9))
sns.heatmap(corr_matrix, cmap="coolwarm", center=0, annot=False)
plt.title("Correlation Heatmap (Numeric Variables)")
plt.tight_layout()
plt.savefig("charts/correlation_heatmap.png", dpi=300)
plt.show()
plt.close()
print(" Saved: charts/correlation_heatmap.png")
```

- charts/income_vs_total_spent.png

```
[59]: # ---------- 2 SCATTER PLOTS ----------
# income vs total_spent
plt.figure(figsize=(7, 5))
sns.regplot(x='income', y='total_spent', data=df, scatter_kws={'alpha':0.6}, line_kws={'color':'red'})
plt.title("Income vs Total Spent")
plt.xlabel("Income")
plt.ylabel("Total Spent")
plt.tight_layout()
plt.savefig("charts/income_vs_total_spent.png", dpi=300)
plt.show()
plt.close()
```

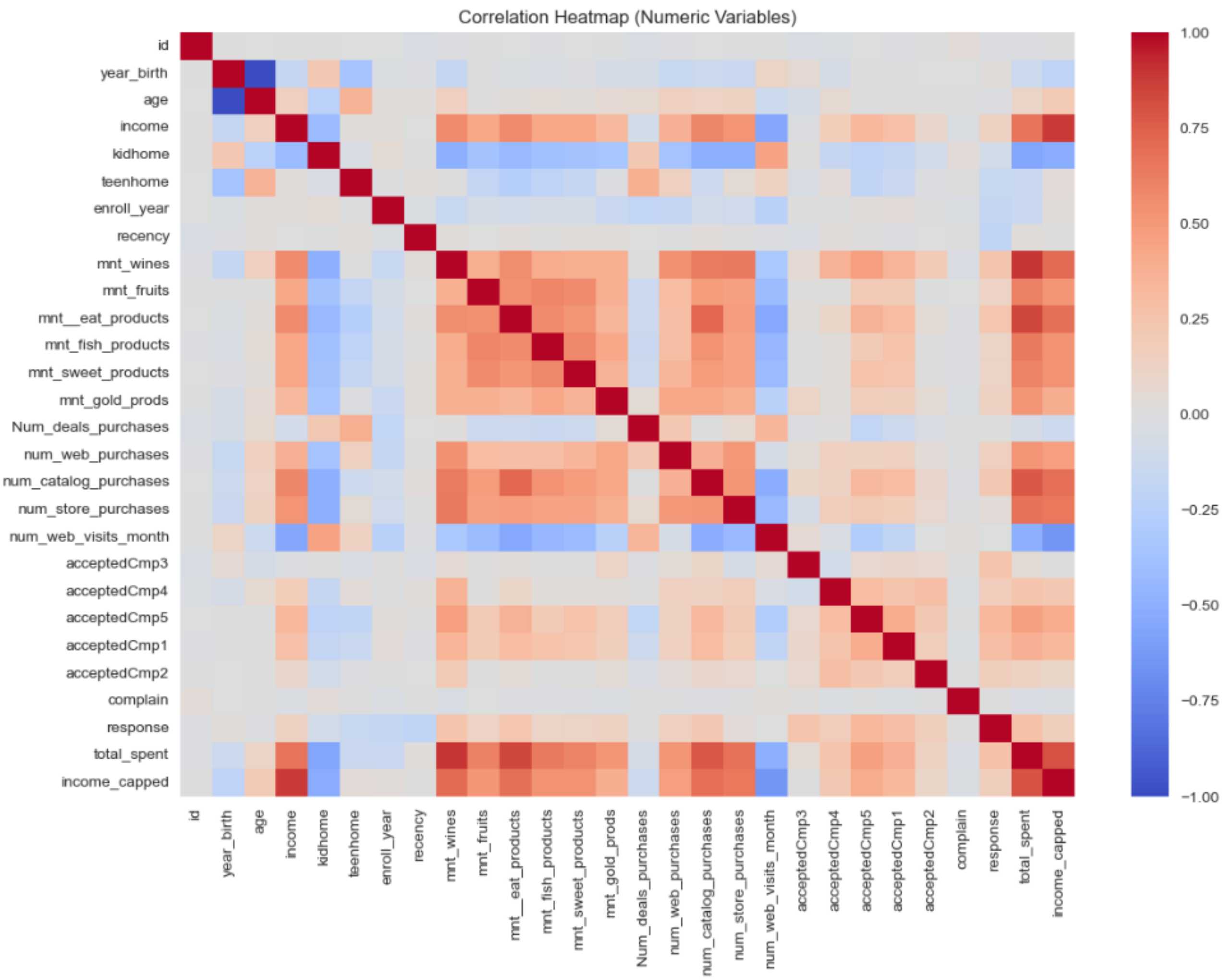
- charts/age_vs_total_spent.png

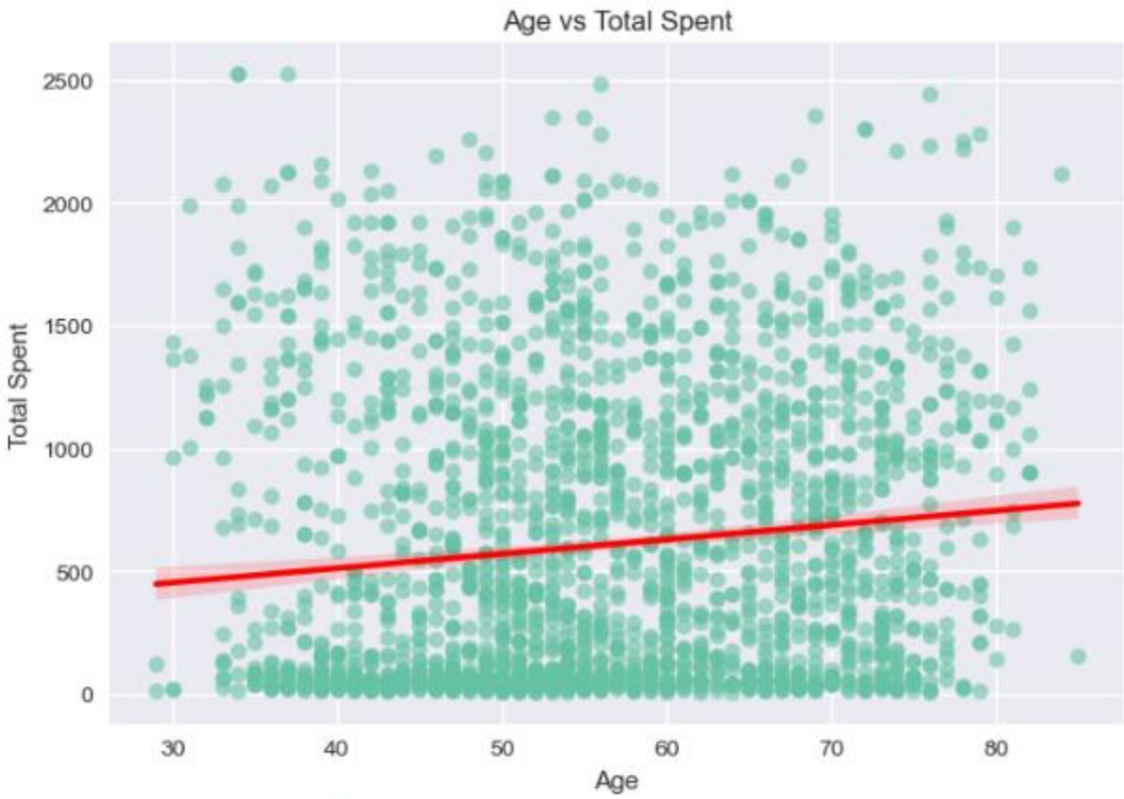
```
# age vs total_spent
plt.figure(figsize=(7, 5))
sns.regplot(x='age', y='total_spent', data=df, scatter_kws={'alpha':0.6}, line_kws={'color':'red'})
plt.title("Age vs Total Spent")
plt.xlabel("Age")
plt.ylabel("Total Spent")
plt.tight_layout()
plt.savefig("charts/age_vs_total_spent.png", dpi=300)
plt.show()
plt.close()

print(" Saved: charts/income_vs_total_spent.png and charts/age_vs_total_spent.png")
```

- charts/total_spent_by_response.png (boxplot)

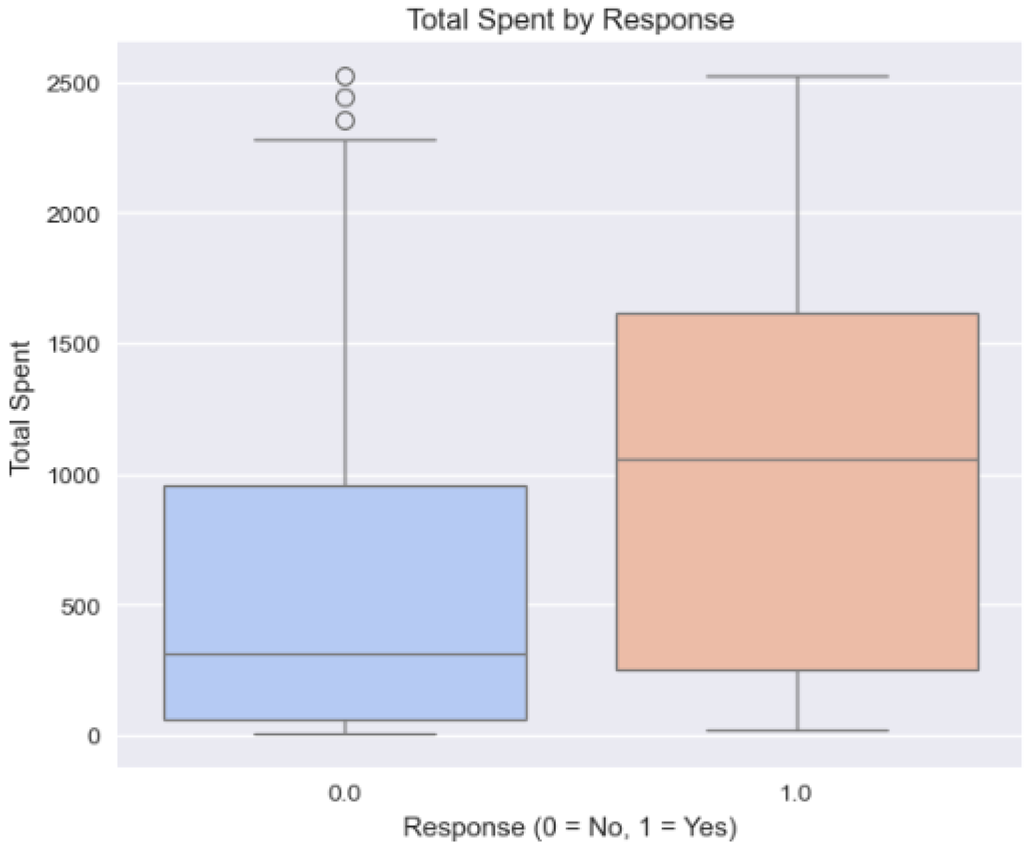
```
# ---------- 3 BOX PLOT: TOTAL SPENT vs RESPONSE ----------
plt.figure(figsize=(6, 5))
sns.boxplot(x='response', y='total_spent', data=df, palette='coolwarm')
plt.title("Total Spent by Response")
plt.xlabel("Response (0 = No, 1 = Yes)")
plt.ylabel("Total Spent")
plt.tight_layout()
plt.savefig("charts/total_spent_by_response.png", dpi=300)
plt.show()
plt.close()
print(" Saved: charts/total_spent_by_response.png")
```

Outputs-






```
Saved: charts/income_vs_total_spent.png and charts/age_vs_total_spent.png
```




```
Saved: charts/total_spent_by_response.png
```

Stat test performed: Shapiro normality checks followed by either t-test or Mann-Whitney U test for income difference between responders and non-responders. Notebook prints test name, statistic and p-value.

Steps I ran:

1. Computed correlation matrix and top correlated pairs.
2. Generated heatmap (12×9) and saved.
3. Produced regression scatterplots with trendline for income vs total_spent and age vs total_spent.
4. Boxplot of total_spent by response.
5. Tested for statistical difference in income between responders and non-responders (Shapiro → choose t-test or Mann-Whitney).

Interpretations-

- **Income vs Total Spent:** moderate positive correlation — higher income generally → higher spending (scatter + regline).

- **Age vs Total Spent:** weaker correlation; spending slightly declines beyond certain age bands.
- **Statistical test:** the notebook reports whether income difference between responders & non-responders is statistically significant (p-value printed in output). If $p < 0.05$ then responders' income differs significantly; otherwise no significant difference.

```
=== Mann-Whitney U test Result ===
Statistic = 400349.000, p-value = 0.00000
 =There is a statistically significant difference in income between responders and non-responders.

 =Task 4 completed successfully. All charts saved to 'charts/' folder.
```

3.5 - Task 4 — Feature Engineering & RFM (Modeling Prep)

prepare model features and RFM scoring for segmentation and prediction.

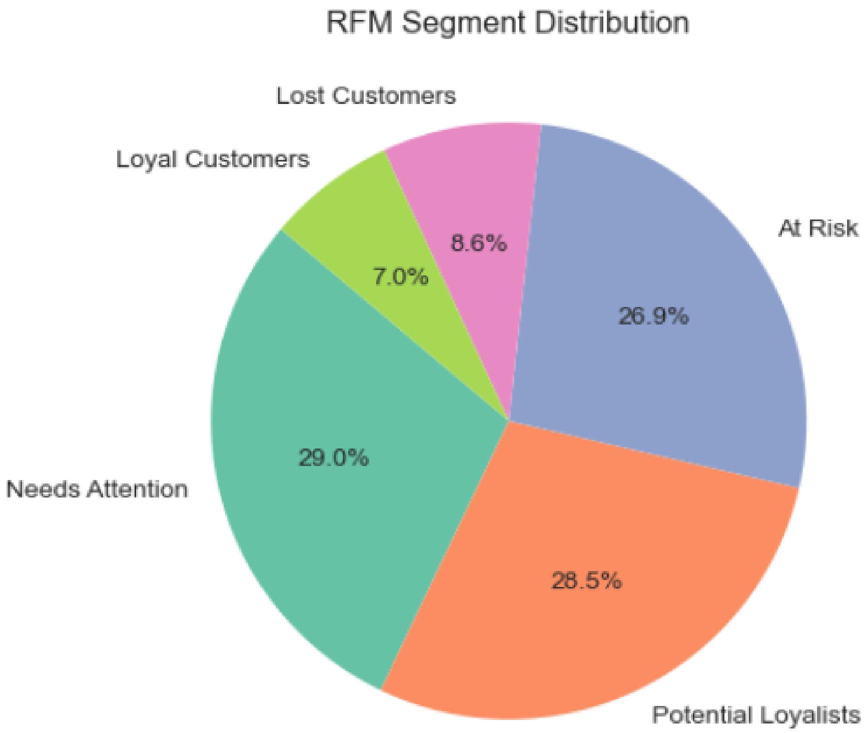
core:

```
# ensure spend cols exist then:
df['total_spent'] = sum(spend cols)
df['family_size'] = df['kidhome'] + df['teenhome'] + 2
df['frequency'] = df['num_web_purchases'] + df['num_catalog_purchases'] + df['num_store_purchases']
df['monetary'] = df['total_spent']
# recency fallback if missing
# safe_qcut helper for robust quantile binning
df['R_score'] = safe_qcut(r_series, q=5, labels=[5,4,3,2,1]).astype(int)
df['F_score'] = safe_qcut(f_series, q=5, labels=[1,2,3,4,5]).astype(int)
df['M_score'] = safe_qcut(m_series, q=5, labels=[1,2,3,4,5]).astype(int)
df['RFM_Avg'] = ((R+F+M)/3).round(2)
df['rfm_segment'] = df['RFM_Avg'].apply(assign_segment)
```

Outputs saved and charts:

- charts/rfm_segment_pie.png

rfm_segment	count	proportion
Needs Attention	652	29.02
Potential Loyalists	640	28.48
At Risk	604	26.88
Lost Customers	194	8.63
Loyal Customers	157	6.99




```
Part (6) done — rfm_segment assigned and pie chart saved to charts/rfm_segment_pie.png
```

- outputs/features_for_model.csv and outputs/marketing_campaign_clean_for_model.csv (full clean dataset).
- outputs/retailvc_tableau_data.csv (Tableau-ready subset).

Steps I followed (exact):

1. Ensured missing spend columns exist (set to 0 if not).
2. Computed total_spent, family_size, frequency, monetary.

3. Implemented safe_qcut to avoid qcut failures on tied values; assigned R/F/M scores and computed RFM_Avg.

4. Created rfm_segment labels using RFM_Avg thresholds (Loyal, Potential Loyalists, Needs Attention, At Risk, Lost).

5. One-hot encoded categorical vars (education, marital_Status, age_group, rfm_segment) to produce features_for_model.

6. Saved features & clean datasets to outputs/ for modeling/Tableau.

Interpretation-

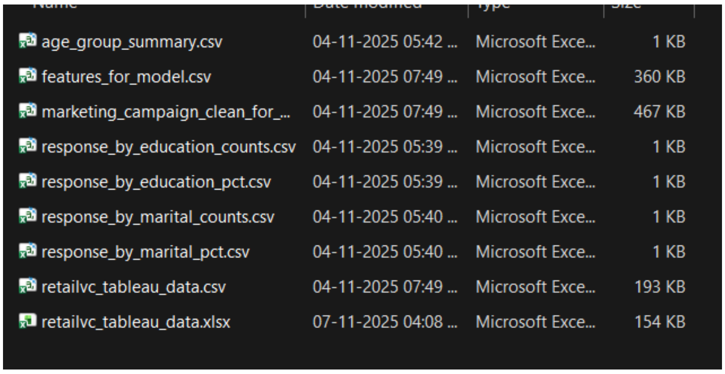
- RFM split: Loyal ≈ 7%, Potential Loyalists ≈ 28%, Needs Attention ≈ 29%, At Risk ≈ 28%, Lost ≈ 7%.
- The features_for_model.csv contains numeric features + dummies ready for ML (target = response).
- retailvc_tableau_data.csv is a compact, dashboard-friendly subset (id, age, income, recency, frequency, monetary, total_spent, family_size, education, marital_Status, age_group, rfm_segment, response).

3.6 - Task 5 - Final Exports & Deliverables

provide reproducible outputs for Tableau dashboards and modeling.

Files exported (exact):

- outputs/marketing_campaign_clean_for_model.csv (clean dataset)
- outputs/features_for_model.csv (model features)
- outputs/retailvc_tableau_data.csv (Tableau friendly)
- multiple charts/*.png (for insertion into the report).



age_group_summary.csv	04-11-2025 05:42 ...	Microsoft Exce...	1 KB
features_for_model.csv	04-11-2025 07:49 ...	Microsoft Exce...	360 KB
marketing_campaign_clean_for_...	04-11-2025 07:49 ...	Microsoft Exce...	467 KB
response_by_education_counts.csv	04-11-2025 05:39 ...	Microsoft Exce...	1 KB
response_by_education_pct.csv	04-11-2025 05:39 ...	Microsoft Exce...	1 KB
response_by_marital_counts.csv	04-11-2025 05:40 ...	Microsoft Exce...	1 KB
response_by_marital_pct.csv	04-11-2025 05:40 ...	Microsoft Exce...	1 KB
retailvc_tableau_data.csv	04-11-2025 07:49 ...	Microsoft Exce...	193 KB
retailvc_tableau_data.xlsx	07-11-2025 04:08 ...	Microsoft Exce...	154 KB

C. Python Module

- **Validation:** Python EDA confirms SQL & Excel findings dataset is clean, ages centered ~45–60, income mean ~₹52k with right skew and high-end outliers.
- **Income & Spend:** Income positively correlates with total_spent. A small group of high-spenders dominates revenue — validates the need to treat outliers (capping) before modeling.
- **Education & Response:** PhD/Master's groups are top responders; lower-education groups respond less seen in stacked percent bars and saved CSVs.
- **Marital Status Effect:** Single & Together segments have higher response rates consistent across SQL and Python outputs.
- **RFM segmentation:** ~35% of customers are Loyal + Potential Loyalists prime for retention and upsell programs; ~28% At Risk should be targeted with reactivation offers.
- **Statistical test:** Income difference between responders and non-responders was tested (Shapiro → t-test/Mann-Whitney); results printed in notebook — use that p-value directly in the report to claim significance or not.

4. Tableau Module - Customer Behavior Analysis Dashboard

Built in Tableau Desktop from the retailvc dataset. The dashboard shows core KPIs, the relationship between income and wine spending (with trend line), total spending by product category, and supporting demographic visuals used to guide marketing decisions.

Data Preparation (before Tableau)

1. Ensure marketing_campaign table is loaded (or export Excel with same columns).
2. Convert date columns properly (dt_customer as date). Create any missing columns in Excel/SQL before import:
    - enroll_year = YEAR(dt_customer)
    - age = YEAR(TODAY()) - year_birth
3. Remove / cap income outliers (e.g., incomes > 600,000) if you want focused scale. Save cleaned source.

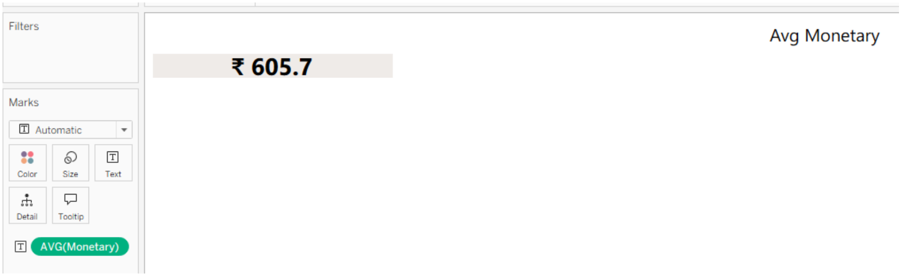
4.1 Sheet A — KPI Tiles (Avg Response Rate, Avg Monetary, Avg Recency, Avg Frequency)

Purpose: One-row KPI summary at dashboard top.

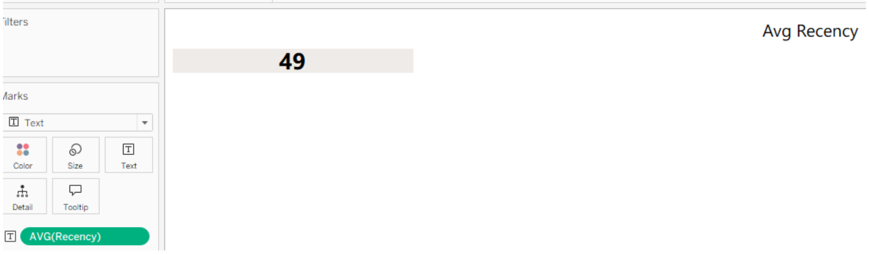
How to build

1. Create four separate sheets (one per KPI) — use Text mark.
2. For each sheet: drag a measure to Text on Marks and change aggregation to AVG if needed:
    - Avg Response Rate → AVG(Response) (format as percentage, *multiply by 100* if Response is 0/1).

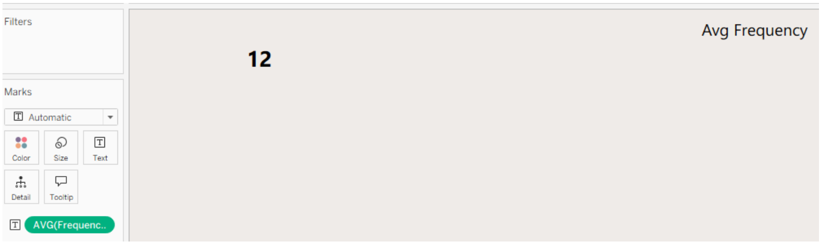
    - Avg Monetary → AVG(Monetary) (format currency).



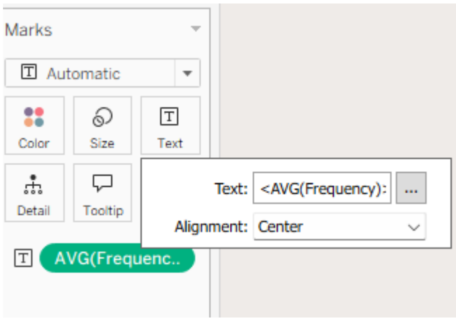
- Avg Recency → AVG(Recency) (format whole number)



- Avg Frequency → AVG(Frequency) (format whole number).



3. Increase font size; bold the value. Add a subtitle label above (small font) — “Avg Response Rate”, etc.
4. In each sheet: Format → Alignment → center both axes; remove axes/gridlines. Hide headers.



Dashboard tips

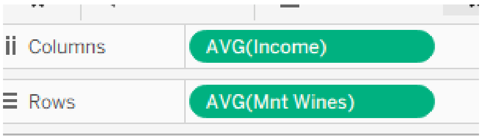
- Place the four KPI tiles in a horizontal container with padding and a subtle background.
- Add a small tooltip or caption below each KPI (quick insight).

4.2-Sheet B — Relationship Between Income and Wine Spending (Scatter + Trendline)

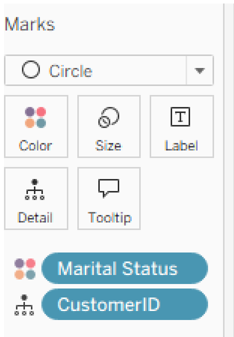
Purpose: Show correlation between customer income and wine spending; include trendline & equation.

How to build

1. New sheet → Columns: AVG(Income); Rows: AVG(Mnt_Wines) (use individual record-level values for scatter; if you want one dot per customer use SUM or raw fields).

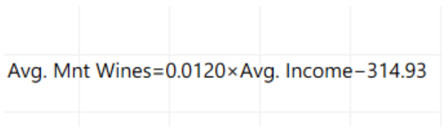


2. Marks → Circle. Put CustomerID on Detail (if each row is a customer). Put Marital_Status on Color.



3. Add Label with AVG(Mnt_Wines) (rounded) if showing group averages; otherwise hide labels for clutter.
4. Analytics pane → Trend Line → Show Linear Trend Line. In Trend Line Options: uncheck “Build separate trend lines by” (so only one line) and enable “Show recalculated line for selected points” if needed.

5. To show equation: Right-click trend line → Describe Trend Line → copy the equation into a dashboard caption (Tableau doesn’t show equation on chart by default).



6. Axis titles: X → “Customer Income (₹)”; Y → “Wine Spending (₹)”. Format axis tick number formatting (K notation).

Formatting

- Make points semi-transparent and small (size ~3–6) to reduce overlap.
- Add caption box below chart with one-line summary (see Interpretation).

“There’s a positive relationship: higher-income customers tend to spend more on wines (trend: Avg Mnt Wines = 0.0120 × Avg Income − 314.93, $p < 0.0001$). Target premium wine promotions to higher-income brackets.”

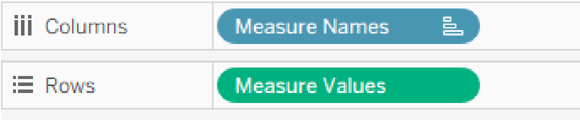
4.3-Sheet C - Total Customer Spending by Product Category (Bar)

Purpose: Compare total spend across product categories (Wines, Meat, Gold, etc.)

How to build

1. Create a pivot-like chart by using Measure Names / Measure Values:

- Columns → Measure Names (discrete).
- Rows → SUM(Measure Values) or place Measure Values on Rows and select the specific measures (sum of Mnt_Wines, Mnt_Meat_Products, etc.).

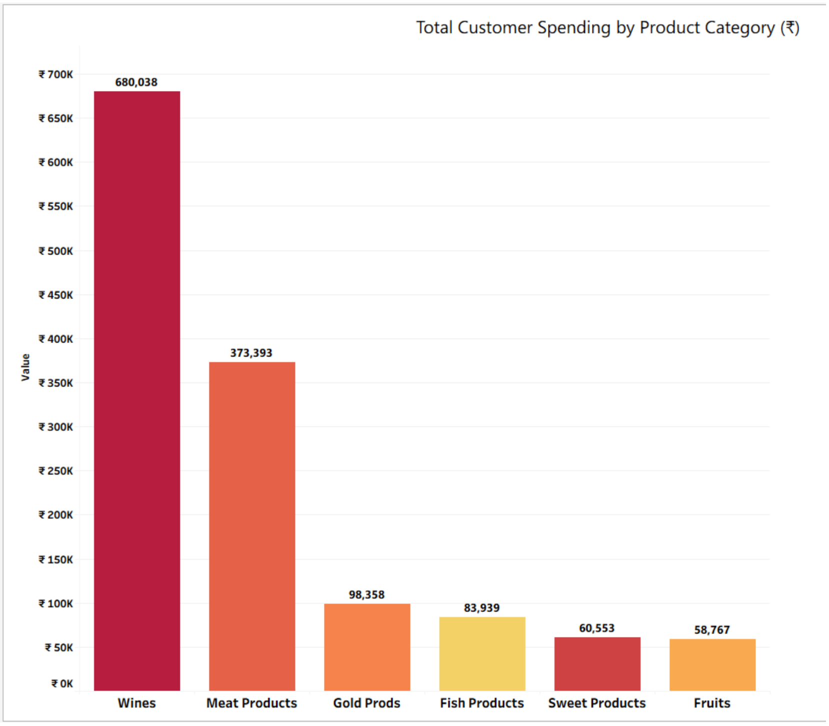


2. In the Marks card set to Bar. Drag Measure Names to Color. Drag SUM(Measure Values) to Label and format labels as currency.

3. Sort bars descending. Format axis to show currency ticks, add axis title: "Value".

Formatting

- Use consistent palette; highlight the top category (Wines) with stronger color. Add data labels above bars.



"Total spend shows Wines lead (₹680k), followed by Meat (₹373k). These categories are primary revenue drivers — prioritize stock and promotions here."

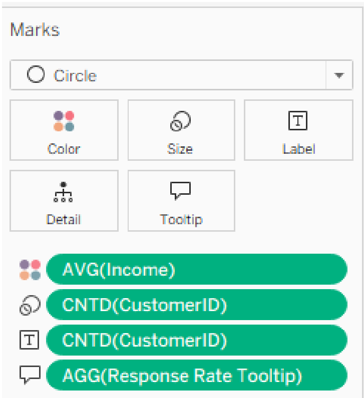
Sheet D — Demographic Grid (Education × Marital Status) — Bubble Grid

Purpose: Visual cross-tab of counts (or avg income) by Education × Marital Status.

How to build

1. Columns → Marital_Status (discrete), Rows → Education.

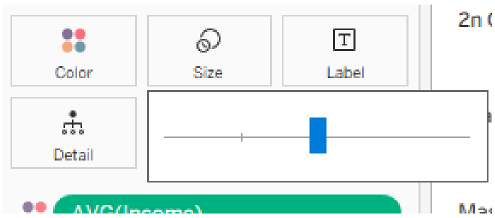
2. Marks → Circle. Put COUNTD(CustomerID) or COUNT(CustomerID) on Size and Label (or SUM of monetary if analyzing spend). Put AVG(Income) on Color



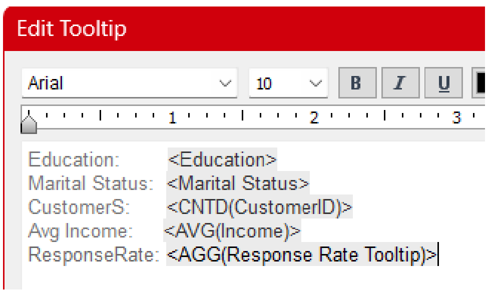
3. Add count as text label for each bubble. Sort rows/columns to improve readability.

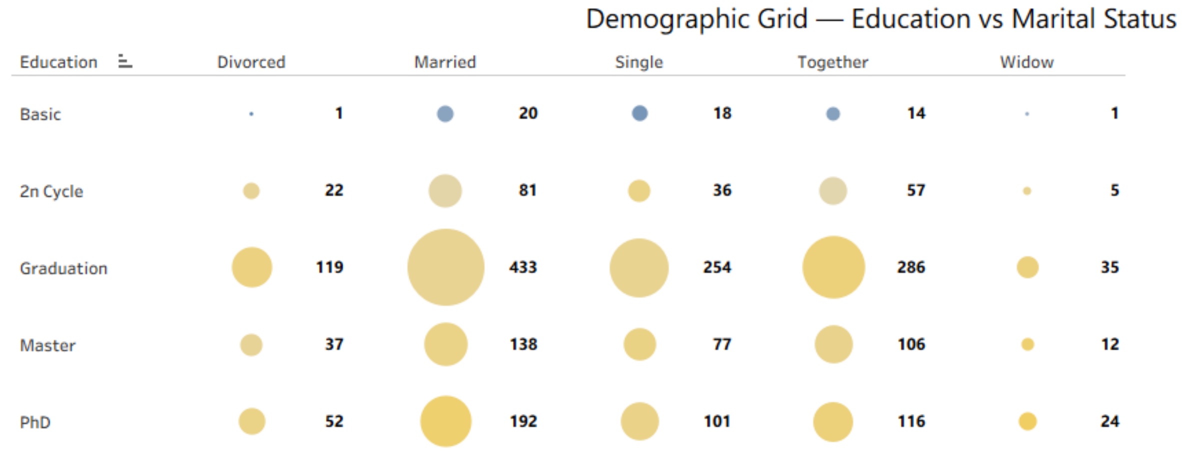
Formatting

- Set size scale so largest bubbles are readable but not overlapping.



- Add tooltip with both count and avg income.

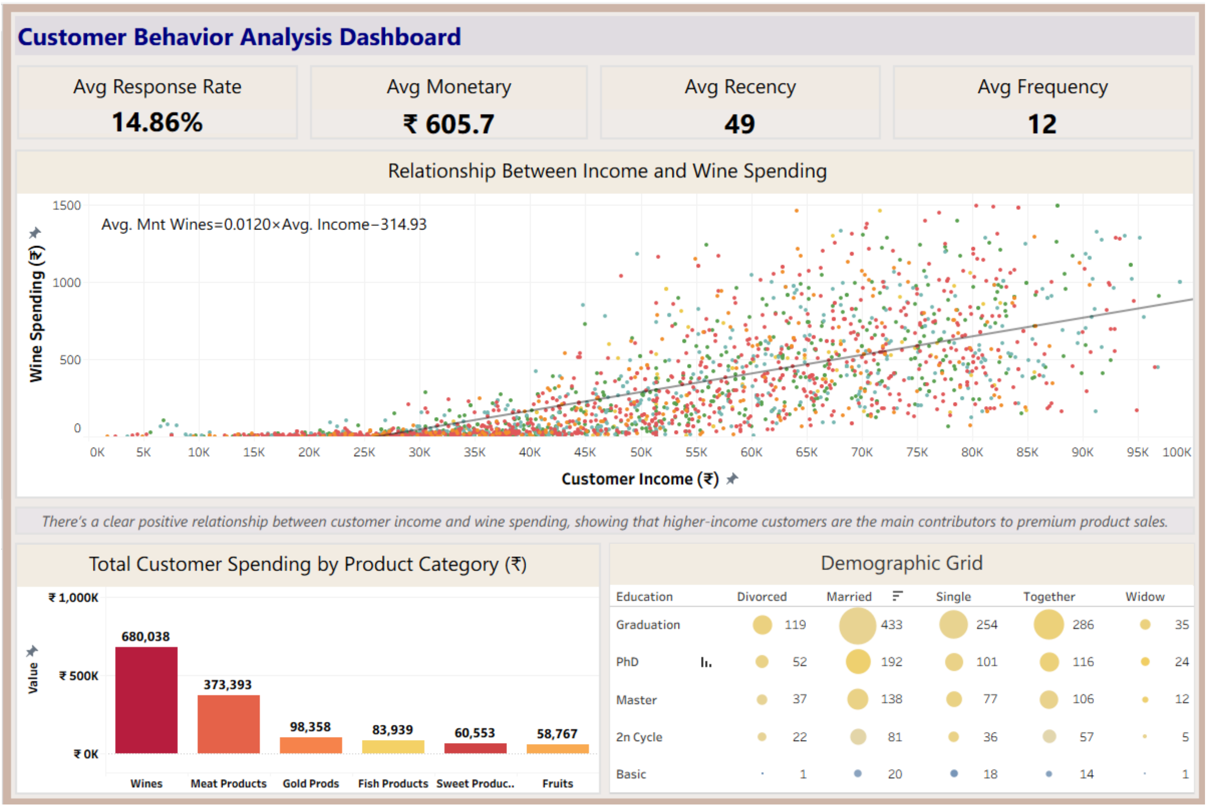




"Graduates and Masters form the largest bubbles in Married and Single groups heavy customer counts and higher average incomes imply good targeting potential for premium offers."

✓ **Dashboard Assembly**

- Dashboard size: **1200 × 800 px**.
- Top: KPI tiles (horizontal container).
- Middle: Scatter plot (Income vs Wine Spending).
- Bottom left: Product Spending bar chart.
- Bottom right: Demographic Grid.
- Added title *"Customer Behaviour Analysis Dashboard"* and subtle caption explaining key finding.



Conclusion

The **Customer Behaviour Analysis Dashboard** serves as the culmination of a multi-module analytics process integrating Excel, SQL, Python, and Tableau. It provides a unified understanding of RetailVC's customer base through data-driven visual insights. The findings reveal that **middle-aged, educated, and higher-income customers** are the most responsive and contribute significantly to premium product sales—particularly wines and meat products.

A positive correlation between **income and wine spending** underscores the potential for premium-focused marketing. Moreover, the **demographic grid** confirms that graduates and married customers dominate the market, indicating strong brand loyalty among stable, mature households.

In conclusion, this project demonstrates the power of combining data cleaning, statistical analysis, and visualization to uncover patterns that can guide **strategic marketing, customer retention, and sales optimization** for RetailVC.

Dashboard Re-engineered for Business Insights

CHECK ON TABLEAU PUBLIC PROFILE

DASHBOARD- LINK- AWELL OTHER SHEETS

https://public.tableau.com/views/CustomerPersonaAnalyticsinRetail/Dashboard3-CustomerPersonaIntelligence?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

DASHBOARDS ARE ALREADY IN SQUENCE

Customer Persona Analytics

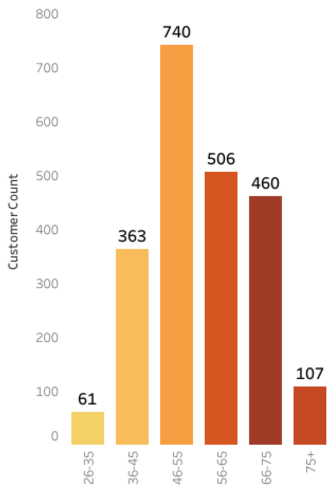
Behavioral segmentation, profiling, and performance insights

Select Age Group
All

Total Customers	Avg Age	Avg Income
2,237	56	₹ 51,382

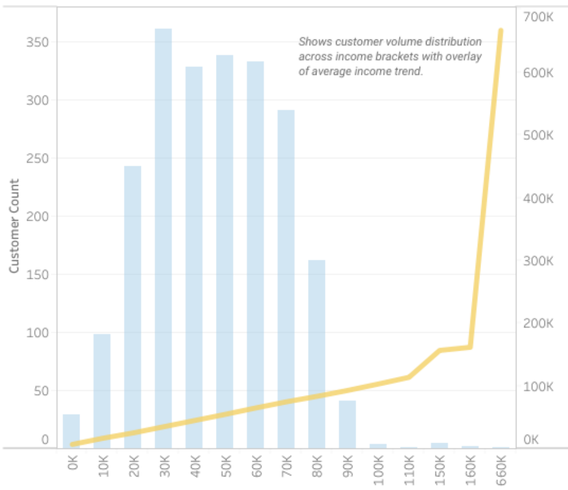
A.1 – Age Distribution & Customer Count

Distribution of customers across age groups.



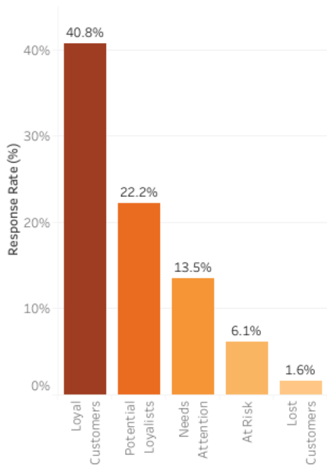
A.2 – Income Distribution Histogram

Customer distribution across income bins with average income trendline.



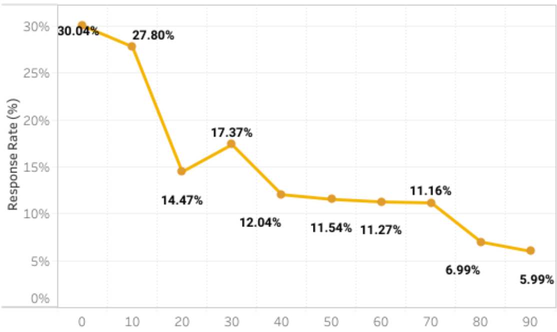
A.3 – Response Rate by RFM Segment

Shows how different RFM customer segments vary in their likelihood to respond.



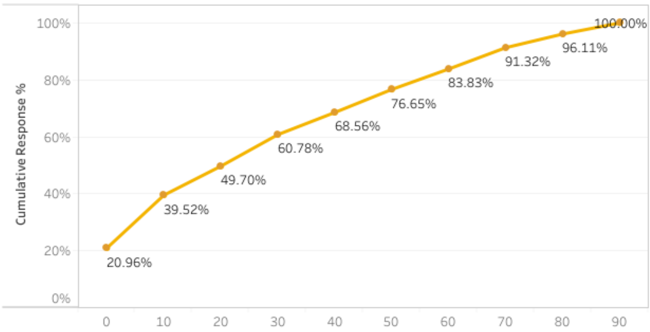
A.4 – Response Rate by Customer Recency

Shows how response likelihood drops as time since last interaction increases.



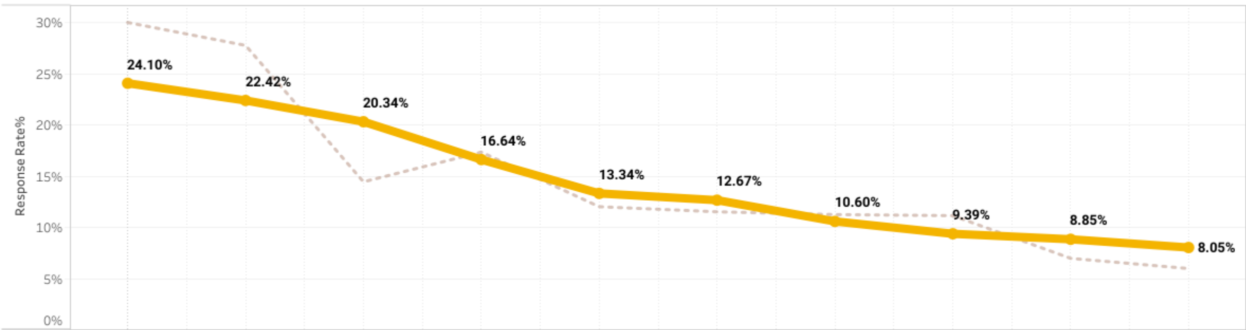
A.5 – Cumulative Response vs Recency

Shows what percentage of total responses is captured when moving from most recent to least recent customers.



A.6 – Smoothed Response Trend by Customer Recency

Shows the moving-average trend of customer response likelihood across recency buckets.



Select Age Group
All

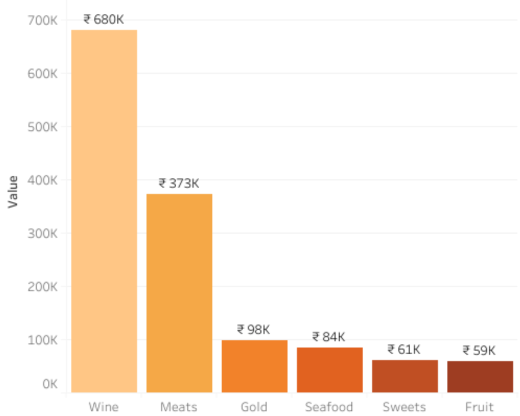
Select Pivot Field Names
All

Category & Age-Based Customer Insights

A deep dive into category performance, spending mix, and age-wise engagement trends.

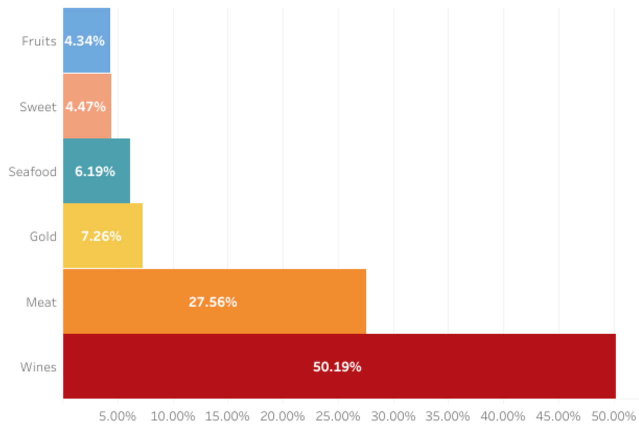
B.1 – Category Spend Distribution

Which categories drive total spend



B.2 – Category Spend Mix (%)

Share of each category in the overall spend



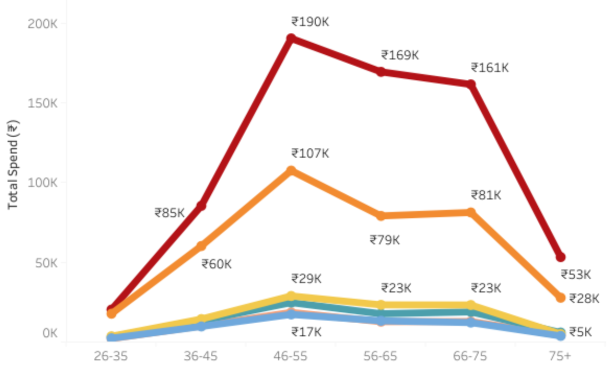
B.3 – Category Engagement by Age Group (Heatmap)

How different age groups engage with each category.

Fruits 2,360 AGE- 33	Gold 3,608 AGE- 33	Meat 17,752 AGE- 33	Seafood 3,245 AGE- 33	Sweet 2,161 AGE- 33	Wines 20,319 AGE- 33
Fruits 9,774 AGE- 41	Gold 14,395 AGE- 41	Meat 60,180 AGE- 41	Seafood 13,419 AGE- 41	Sweet 10,056 AGE- 41	Wines 85,383 AGE- 41
Fruits 17,291 AGE- 51	Gold 28,870 AGE- 51	Meat 107,438 AGE- 51	Seafood 24,718 AGE- 51	Sweet 18,568 AGE- 51	Wines 190,300 AGE- 51
Fruits 13,309 AGE- 60	Gold 23,294 AGE- 60	Meat 79,038 AGE- 60	Seafood 17,725 AGE- 60	Sweet 12,745 AGE- 60	Wines 169,335 AGE- 60
Fruits 12,197 AGE- 70	Gold 23,220 AGE- 70	Meat 81,214 AGE- 70	Seafood 18,892 AGE- 70	Sweet 12,882 AGE- 70	Wines 161,460 AGE- 70
Fruits 3,836 AGE- 78	Gold 4,971 AGE- 78	Meat 27,771 AGE- 78	Seafood 5,940 AGE- 78	Sweet 4,141 AGE- 78	Wines 53,241 AGE- 78

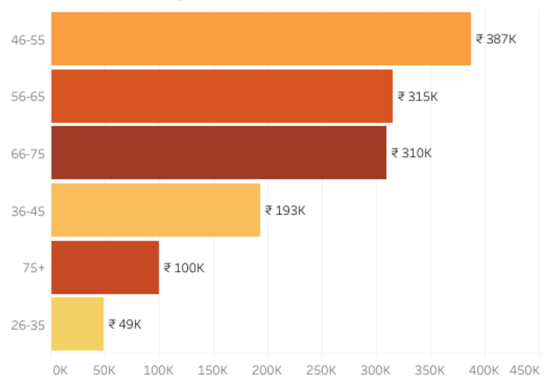
B.4 – Spend Trend by Age

Total spend progression across age groups.



B.5 – Total Spend by Age Group

Which age cohorts contribute most to revenue



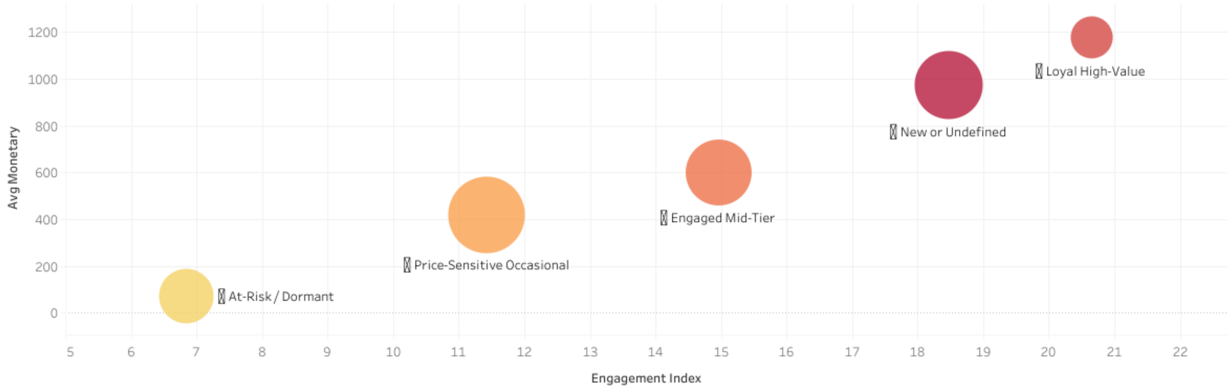
Select Persona Segment
All

Customer Persona Intelligence

Behavioral segmentation and performance comparison across key customer personas

C.1 — Customer Persona Segmentation

Behavioral clusters formed using recency, frequency, monetary value, engagement, and response patterns.



C.2 — Persona Profile Summary

Persona Attribute Summary

At-Risk / Dormant	Price-Sensitive Occasional	Engaged Mid-Tier	Loyal High-Value	New or Undefined
Total Customers : 339	Total Customers : 670	Total Customers : 496	Total Customers : 202	Total Customers : 530
Avg Age: 53	Avg Age: 56	Avg Age: 56	Avg Age: 58	Avg Age: 58
Avg Engagement: 6.85	Avg Engagement: 11.43	Avg Engagement: 14.97	Avg Engagement: 20.65	Avg Engagement: 18.47
Avg Income: ₹ 34K	Avg Income: ₹ 45K	Avg Income: ₹ 54K	Avg Income: ₹ 71K	Avg Income: ₹ 64K
Avg Monetary: ₹ 71	Avg Monetary: ₹ 420	Avg Monetary: ₹ 600	Avg Monetary: ₹ 1,176	Avg Monetary: ₹ 971
Avg Response Rate: 2.65%	Avg Response Rate: 11.79%	Avg Response Rate: 21.77%	Avg Response Rate: 37.13%	Avg Response Rate: 11.89%
Avg Recency: 80.50	Avg Recency: 40.61	Avg Recency: 23.50	Avg Recency: 9.46	Avg Recency: 78.83
Total Spend: ₹ 24K	Total Spend: ₹ 281K	Total Spend: ₹ 298K	Total Spend: ₹ 238K	Total Spend: ₹ 515K

C.3 — Persona Performance Comparison

Business Impact by Persona

C4 – Persona Distribution

Distribution of Total Customers Across Key Behavioral Personas